

Nassau River-St. Johns River Marshes Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

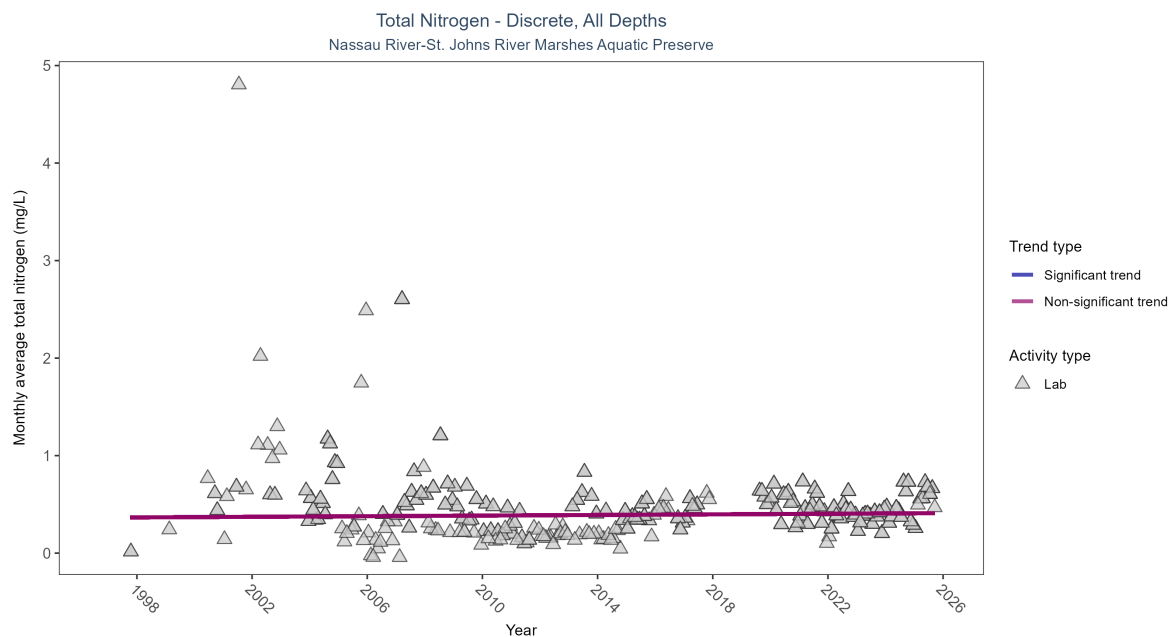


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	1079	27	1997 - 2025	0.419	0.03332	0.36456	0.00156	0.5048

Total nitrogen showed no detectable trend between 1997 and 2025.

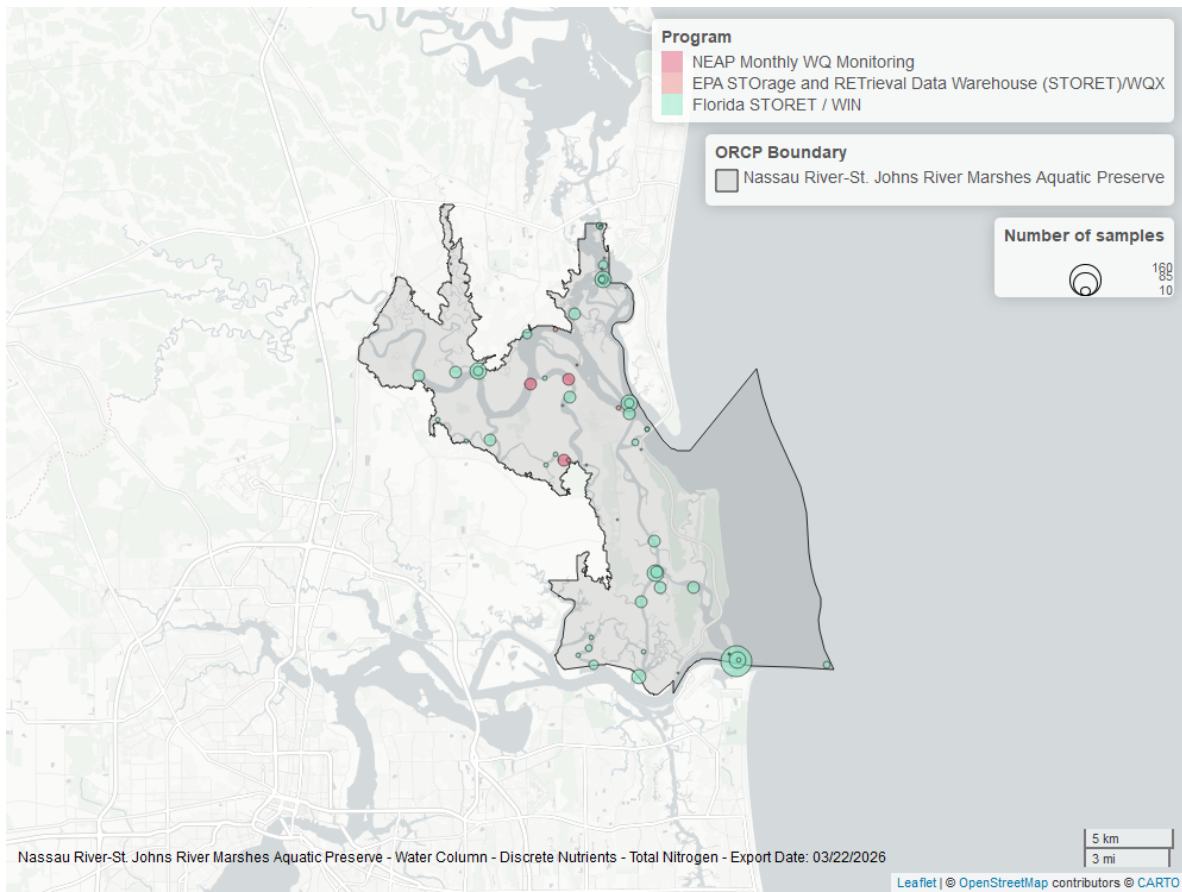


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

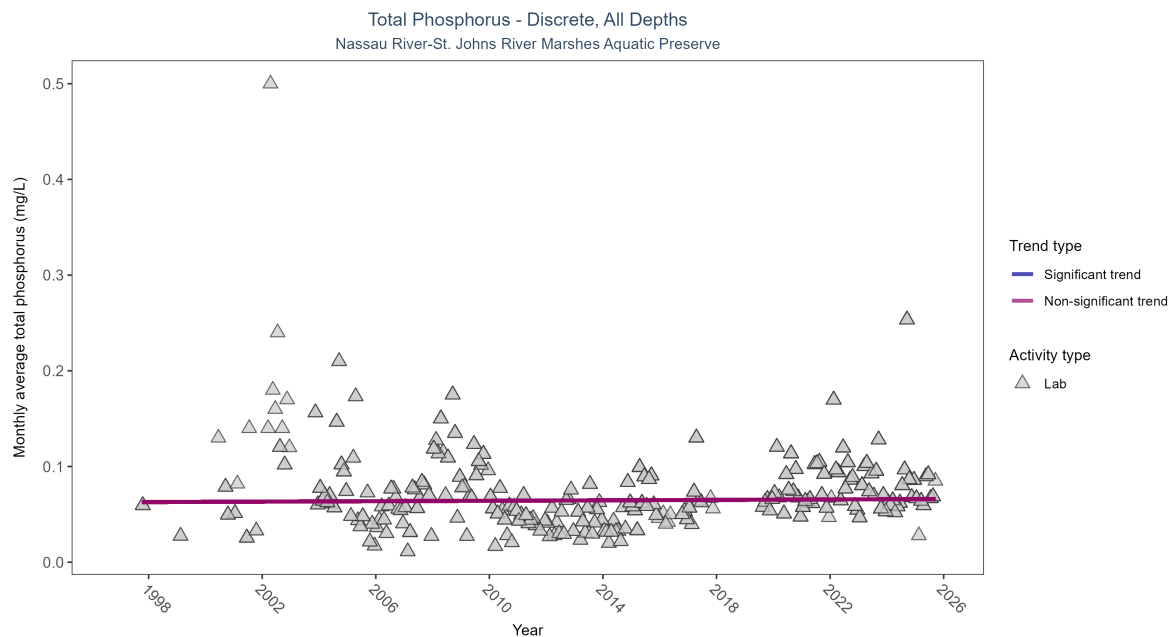


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	1615	27	1997 - 2025	0.067	0.01638	0.06274	0.00011	0.6539

Total phosphorus showed no detectable trend between 1997 and 2025.

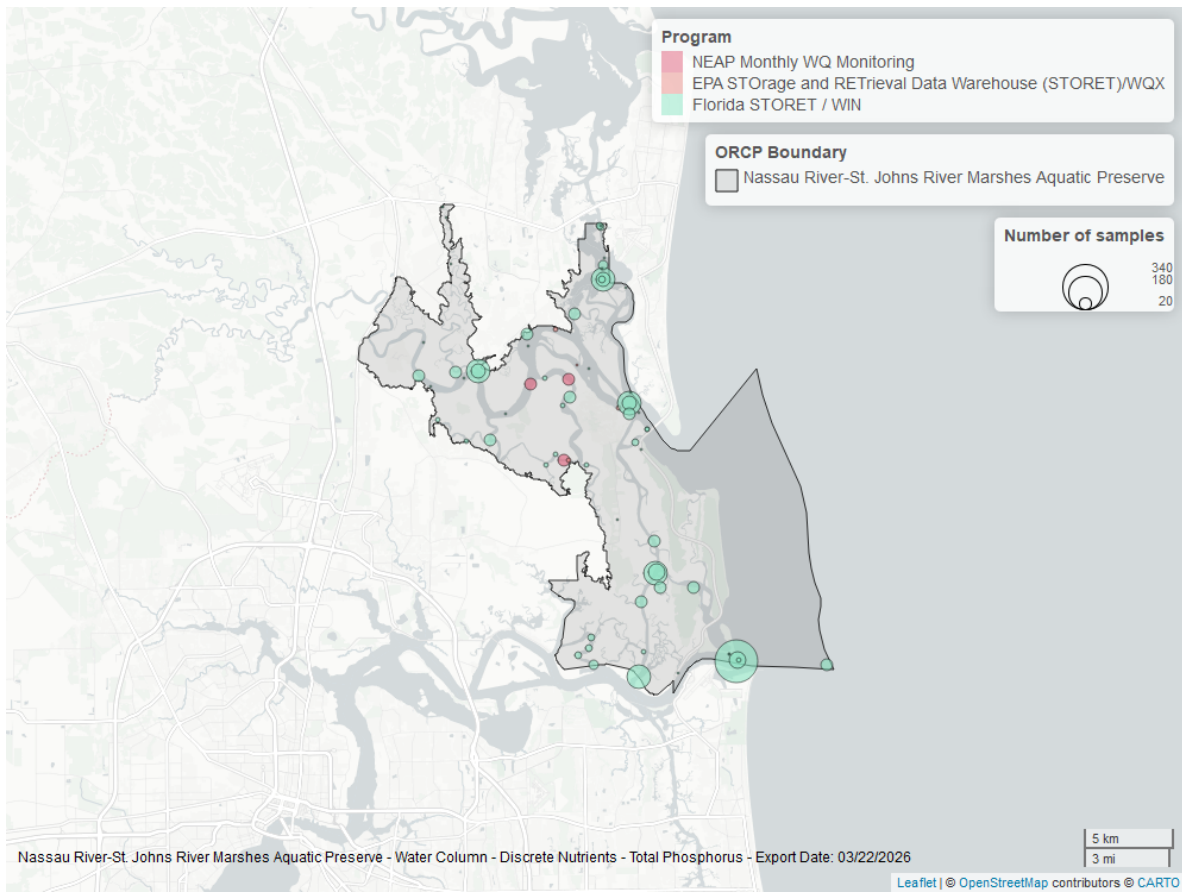


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

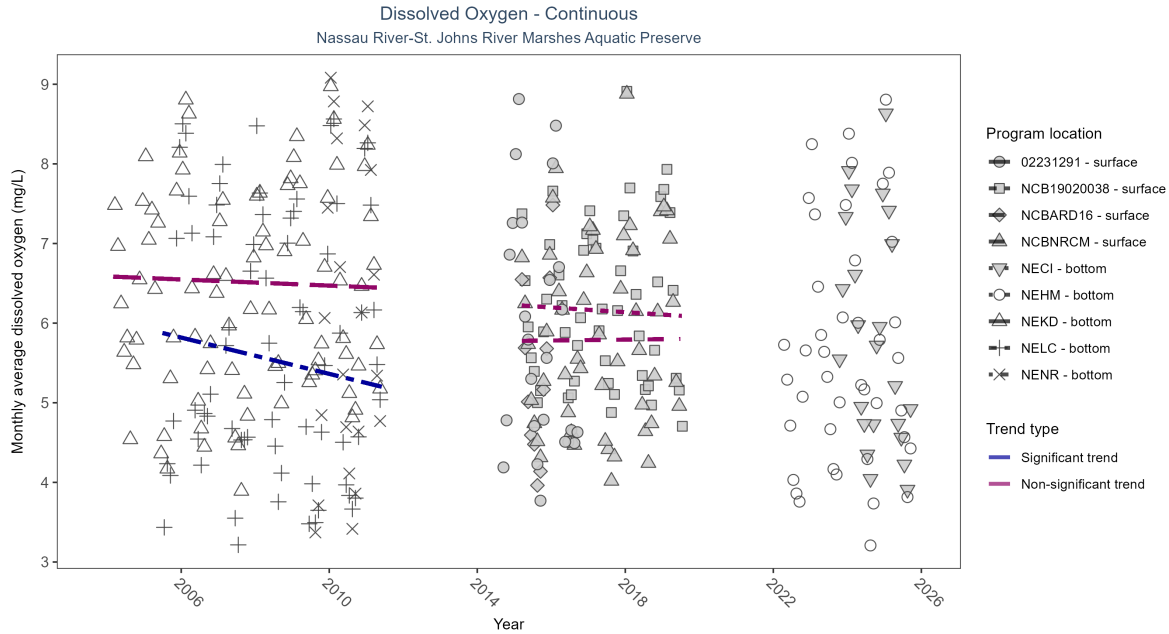


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02231291	Insufficient data to calculate trend	689	3	2014 - 2016	5.70	-	-	-	-
NECI	Insufficient data to calculate trend	62230	3	2023 - 2025	5.80	-	-	-	-
NEHM	Insufficient data to calculate trend	111776	4	2022 - 2025	5.50	-	-	-	-
NEKD	No significant trend	110958	8	2004 - 2011	6.40	-0.04	6.59	-0.02	0.5607
NELC	Significantly decreasing trend	95860	7	2005 - 2011	5.60	-0.34	5.93	-0.11	0.0014
NENR	Insufficient data to calculate trend	31438	3	2009 - 2011	5.80	-	-	-	-
NCB19020038	No significant trend	34476	5	2015 - 2019	6.16	-0.03	6.23	-0.03	0.6108
NCBARD16	Insufficient data to calculate trend	7417	2	2015 - 2016	5.08	-	-	-	-
NCBNRCM	No significant trend	35477	5	2015 - 2019	5.79	0.05	5.78	0.01	0.797

At one program location, monthly average dissolved oxygen decreased by 0.11 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at three locations. There was insufficient data to fit a model for five locations.

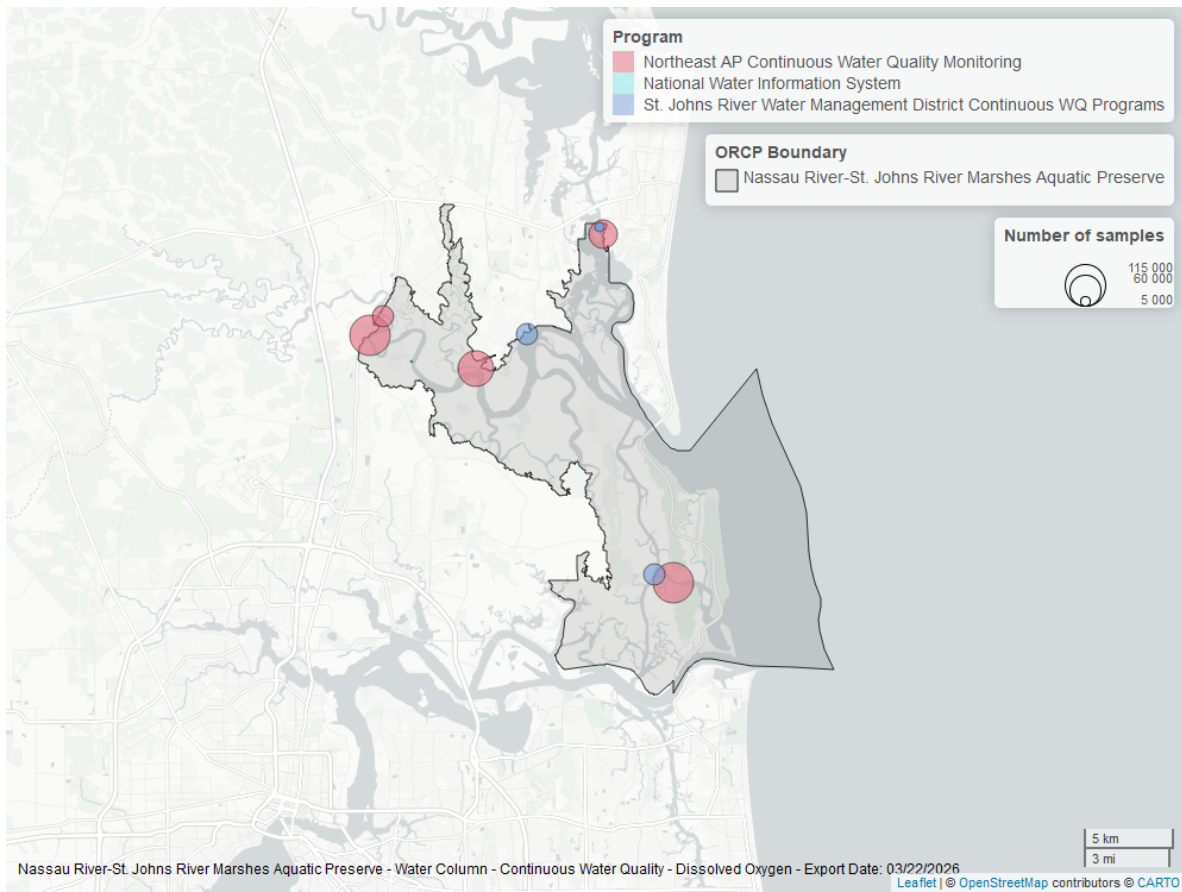


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

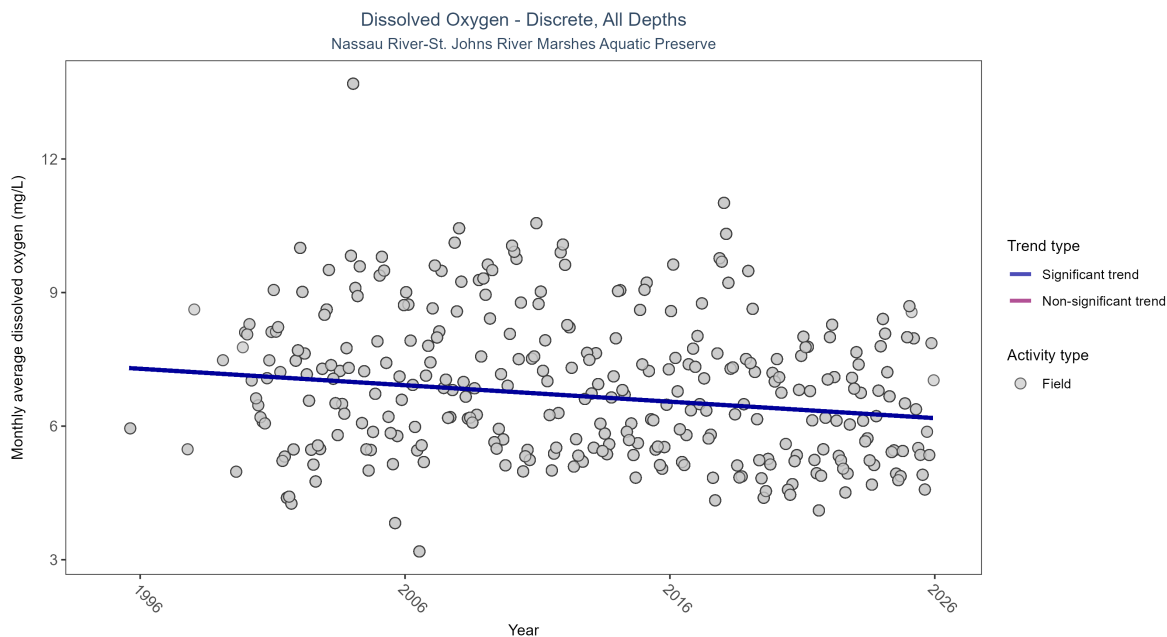


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	22688	30	1995 - 2025	6.8	-0.27269	7.32517	-0.03694	0

Monthly average dissolved oxygen decreased by 0.04 mg/L per year.

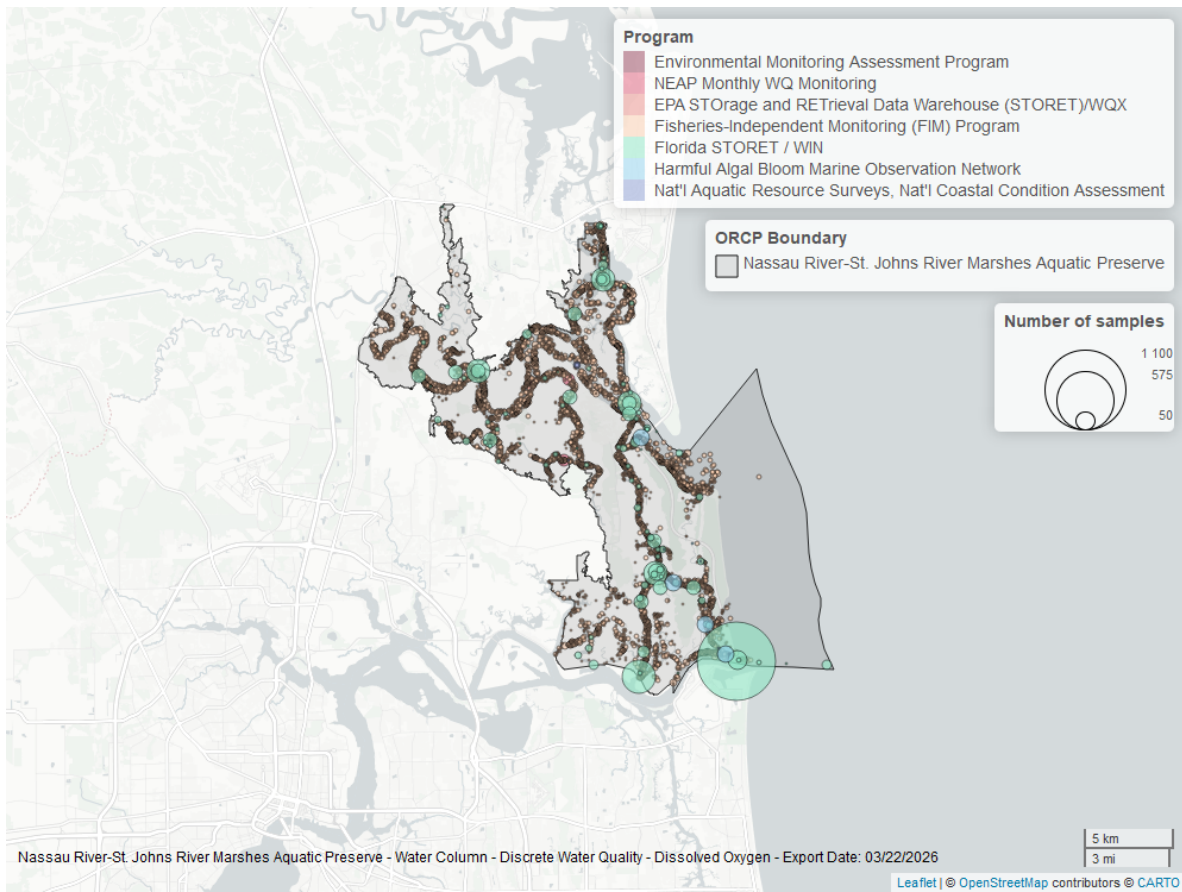


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous



Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data to calculate trend	62577	3	2023 - 2025	80.50	-	-	-	-
NEHM	Insufficient data to calculate trend	114458	4	2022 - 2025	71.50	-	-	-	-
NEKD	No significant trend	110983	8	2004 - 2011	89.10	0	87.49	-0.04	0.9338
NELC	Significantly decreasing trend	95868	7	2005 - 2011	77.90	-0.27	78.93	-1.6	0.0111
NENR	Insufficient data to calculate trend	31438	3	2009 - 2011	73.20	-	-	-	-
NCB19020038	No significant trend	34438	5	2015 - 2019	87.06	-0.16	86.02	-0.6	0.1268
NCBARD16	Insufficient data to calculate trend	7417	2	2015 - 2016	74.44	-	-	-	-
NCBNRCM	No significant trend	35240	5	2015 - 2019	82.06	-0.05	81.1	-0.14	0.797

At one program location, monthly average dissolved oxygen saturation decreased by 1.60% per year. No detectable change in monthly average dissolved oxygen saturation was observed at three locations. There was insufficient data to fit a model for four locations.

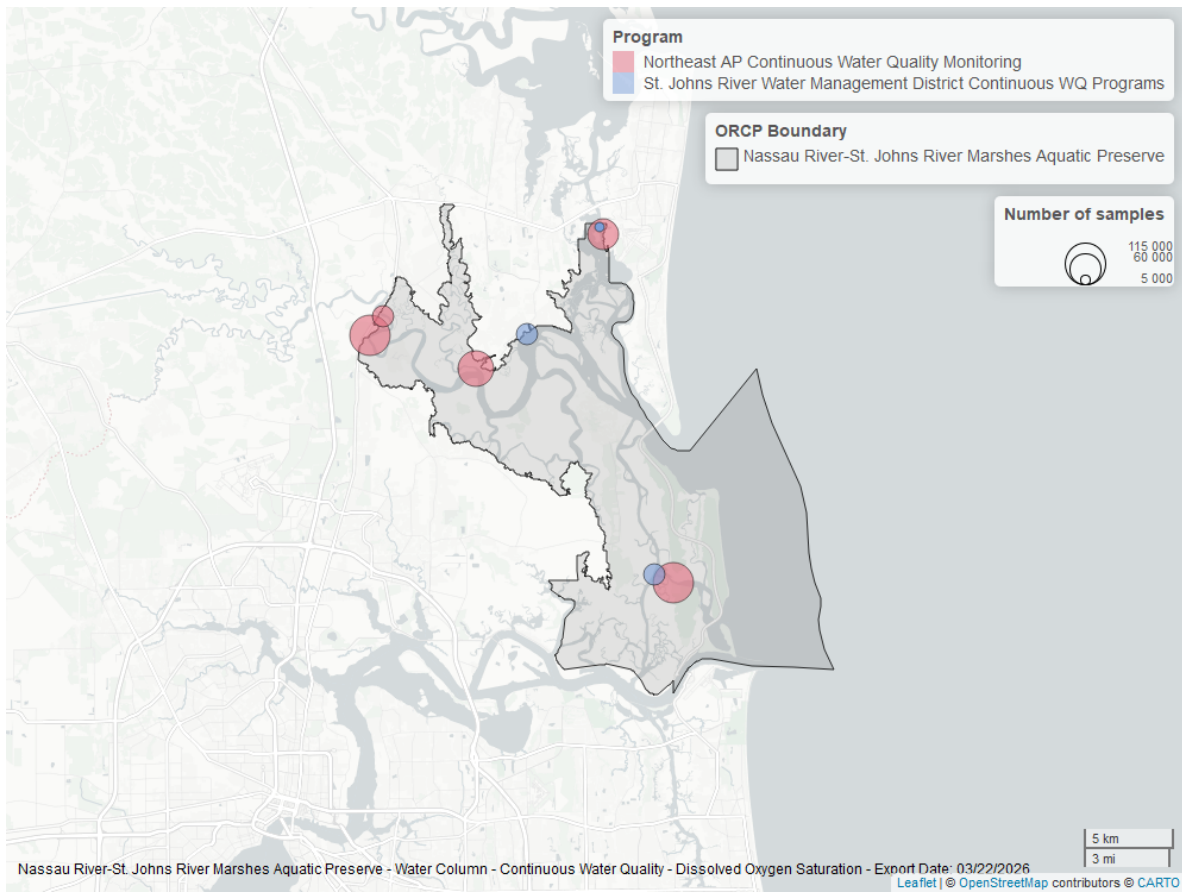


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

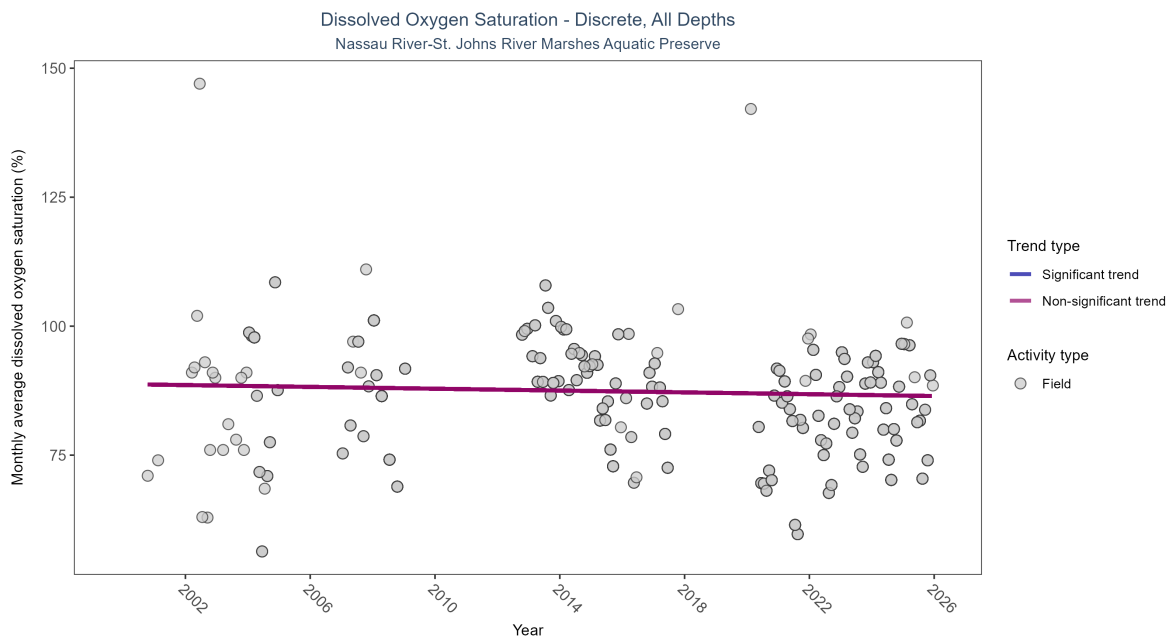


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	1461	20	2000 - 2025	88	-0.08578	88.7588	-0.08849	0.1199

Dissolved oxygen saturation showed no detectable trend between 2000 and 2025.

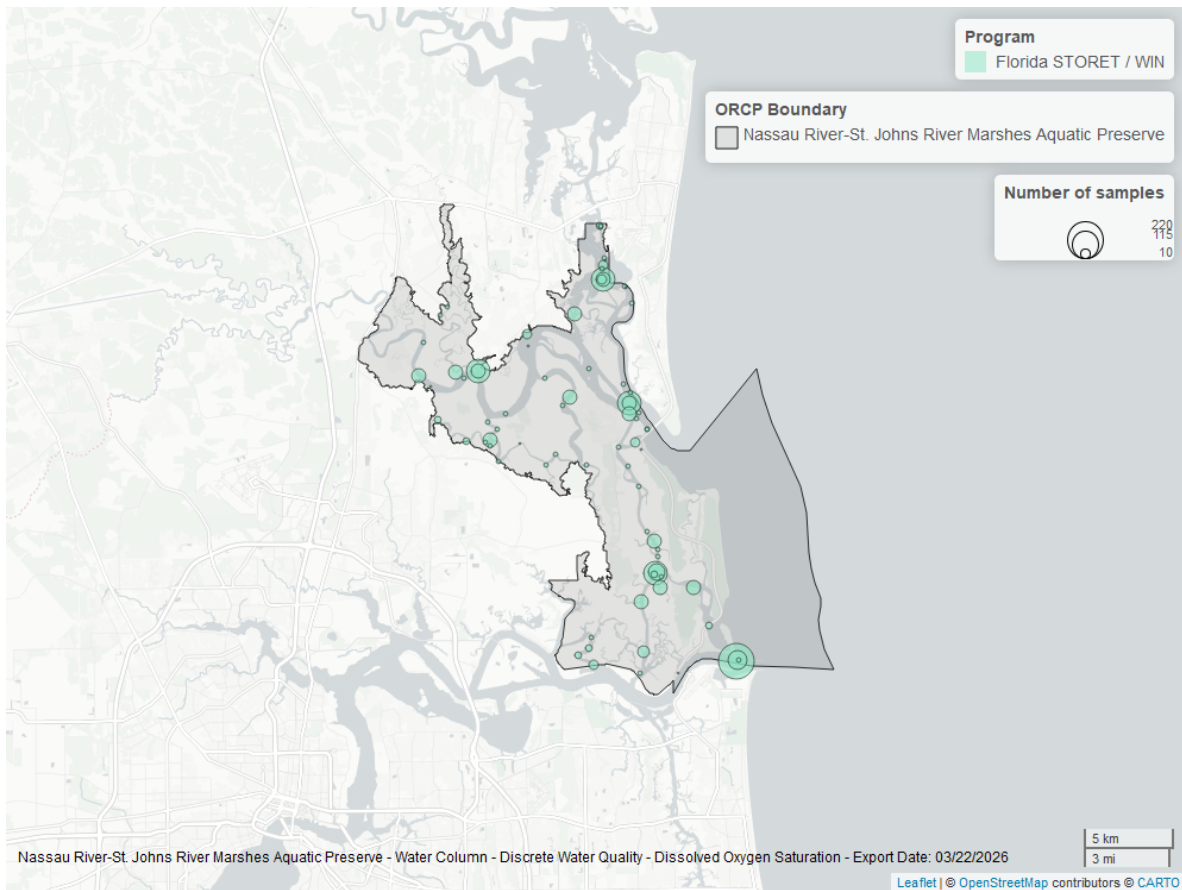


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

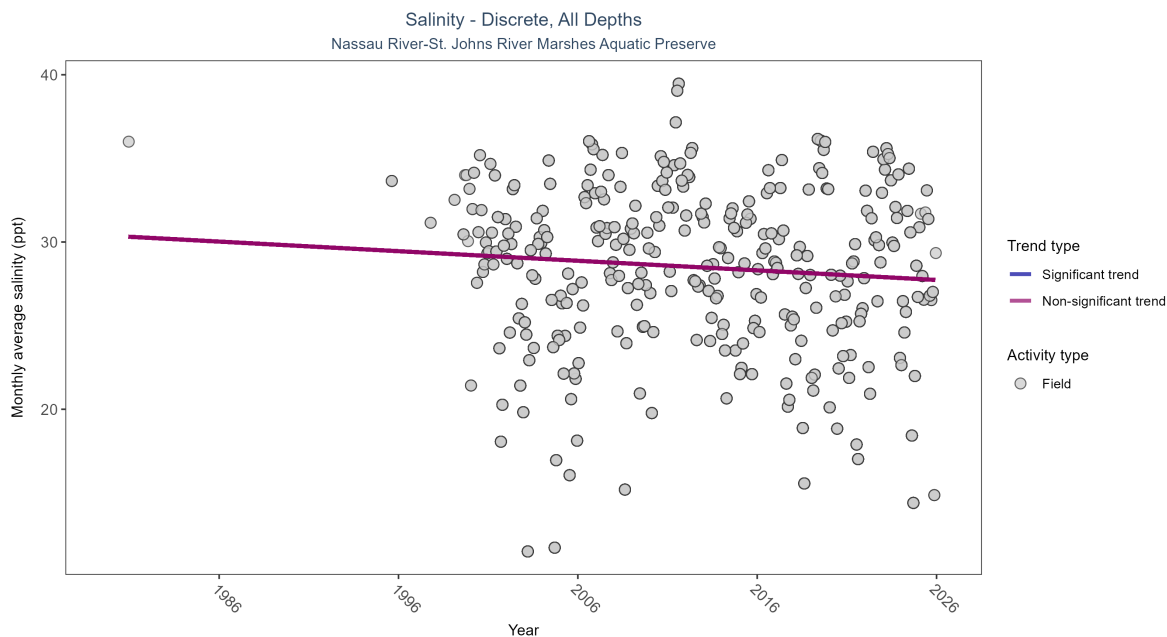


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No significant trend	22797	30	1980 - 2025	30.5	-0.0641	30.36939	-0.05737	0.0985

Salinity showed no detectable trend between 1980 and 2025.

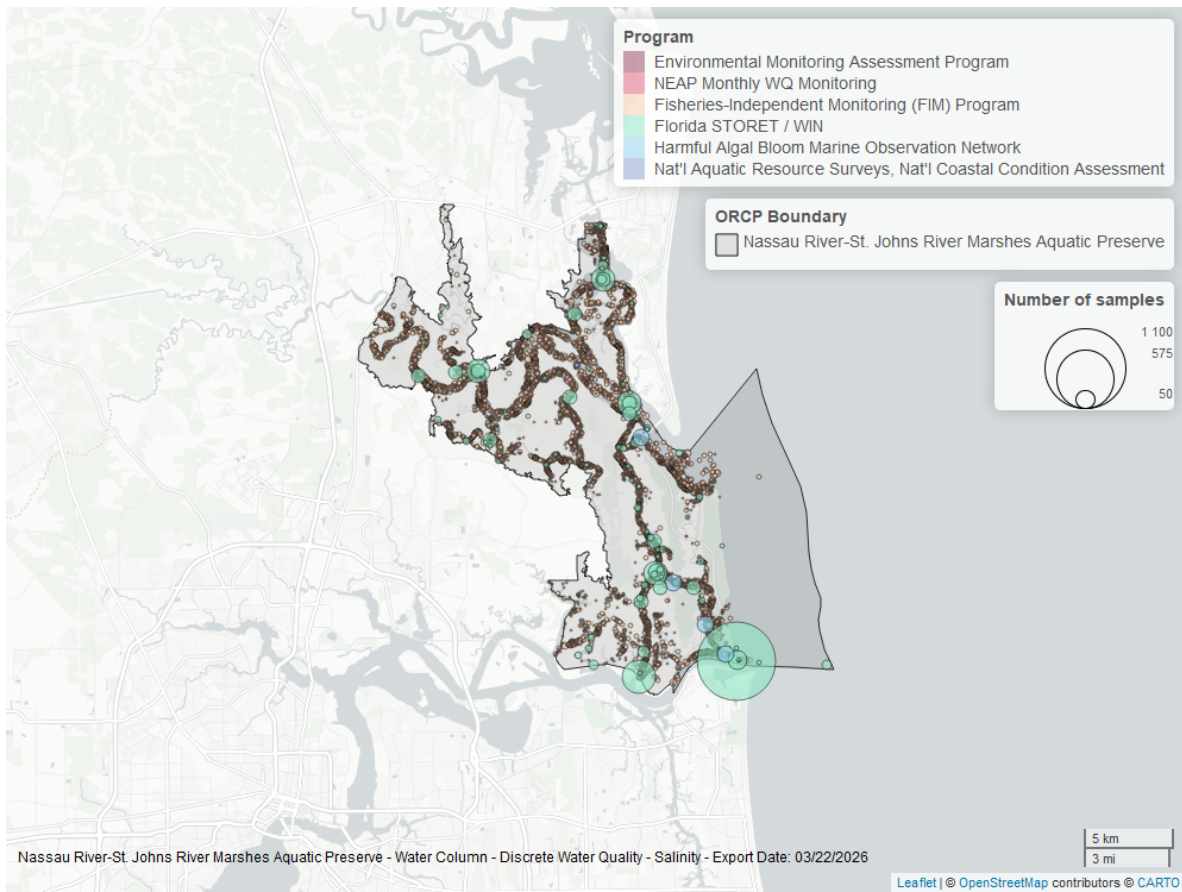


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

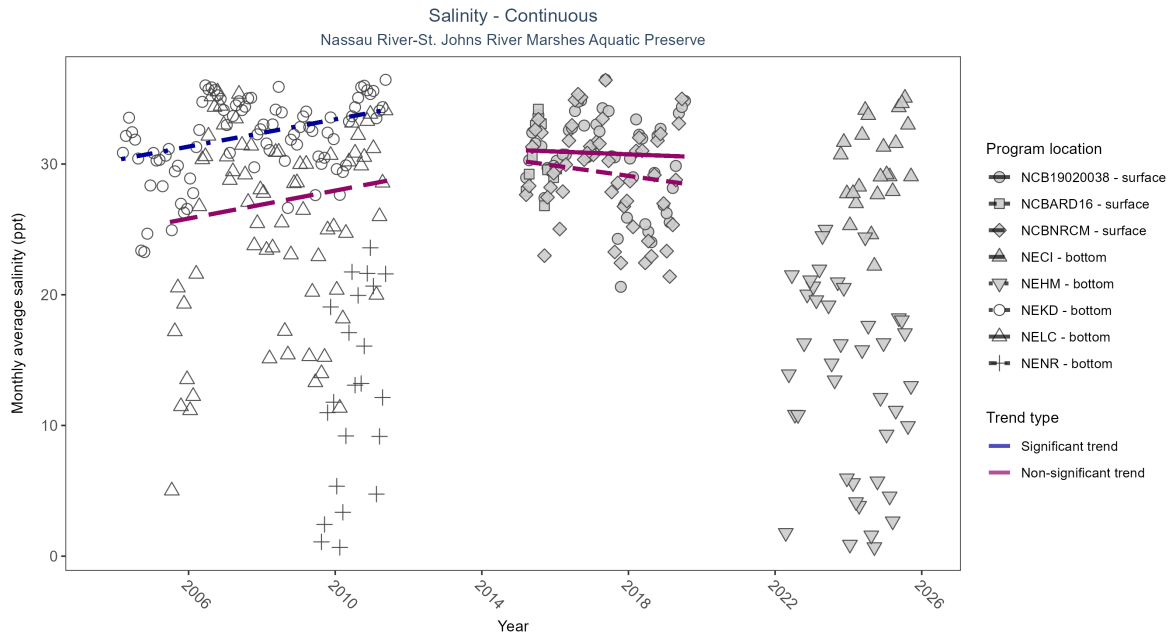


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data to calculate trend	62231	3	2023 - 2025	29.80	-	-	-	-
NEHM	Insufficient data to calculate trend	113173	4	2022 - 2025	14.40	-	-	-	-
NEKD	Significantly increasing trend	118328	8	2004 - 2011	33.20	0.31	30.29	0.52	0.0006
NELC	No significant trend	100339	7	2005 - 2011	27.90	0.09	25.3	0.53	0.4397
NENR	Insufficient data to calculate trend	31438	3	2009 - 2011	12.40	-	-	-	-
NCB19020038	No significant trend	34438	5	2015 - 2019	31.75	-0.06	31.07	-0.11	1
NCBARD16	Insufficient data to calculate trend	7418	2	2015 - 2016	30.07	-	-	-	-
NCBNRCM	No significant trend	35411	5	2015 - 2019	30.29	-0.12	30.26	-0.39	0.5483

At one program location, monthly average salinity increased by 0.52 ppt per year. No detectable change in monthly average salinity was observed at three locations. There was insufficient data to fit a model for four locations.

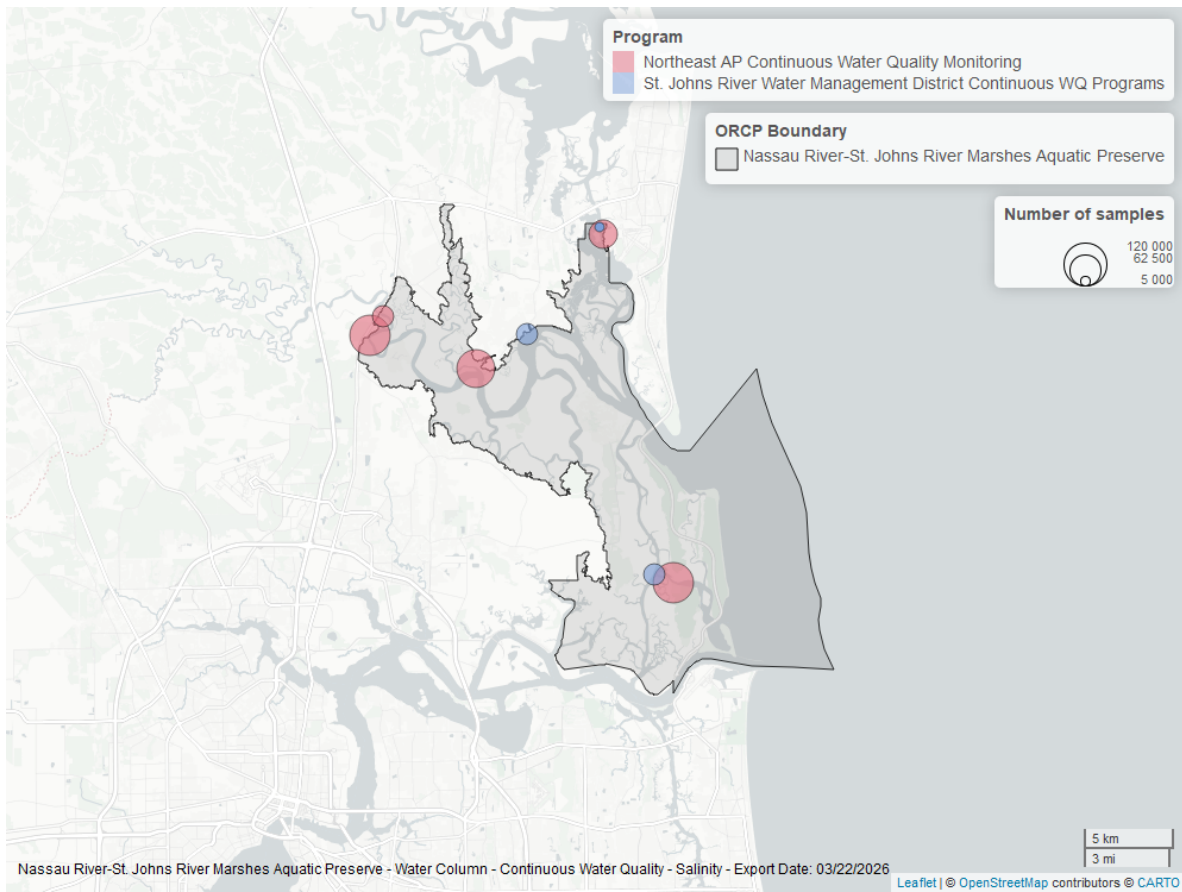


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

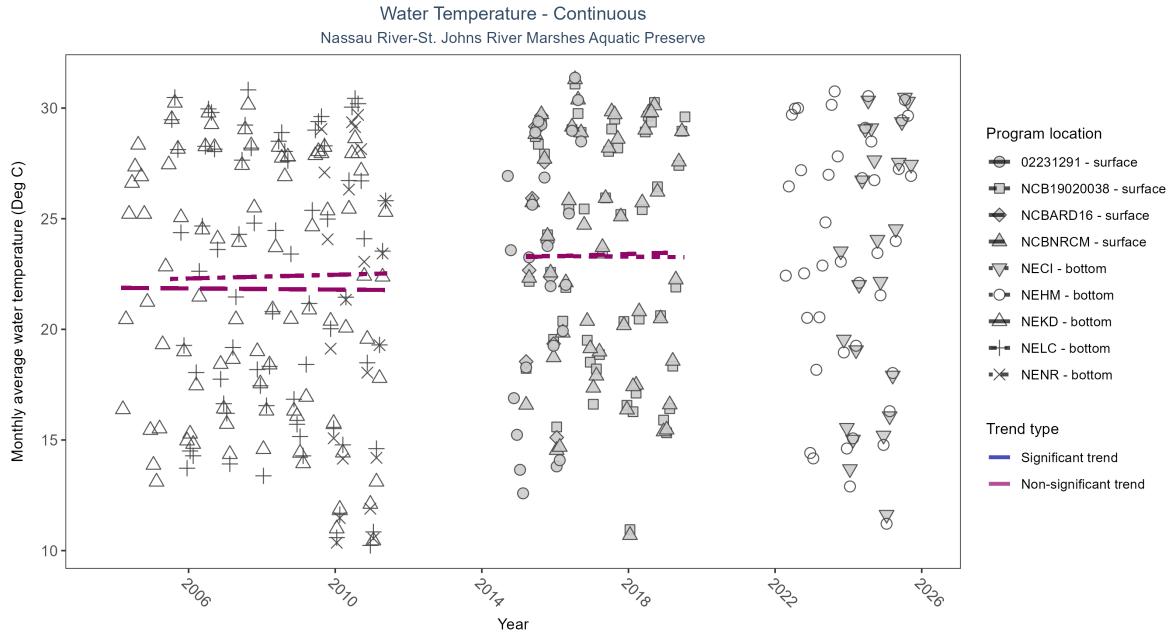


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02231291	Insufficient data to calculate trend	710	3	2014 - 2016	23.50	-	-	-	-
NECI	Insufficient data to calculate trend	62594	3	2023 - 2025	23.10	-	-	-	-
NEHM	Insufficient data to calculate trend	118362	4	2022 - 2025	24.60	-	-	-	-
NEKD	No significant trend	118328	8	2004 - 2011	22.10	-0.04	21.88	-0.01	0.8057
NELC	No significant trend	100343	7	2005 - 2011	22.70	0.05	22.26	0.04	0.7405
NENR	Insufficient data to calculate trend	31438	3	2009 - 2011	22.00	-	-	-	-
NCB19020038	No significant trend	34483	5	2015 - 2019	24.78	-0.03	23.32	-0.01	1
NCBARD16	Insufficient data to calculate trend	7419	2	2015 - 2016	25.08	-	-	-	-
NCBNRCM	No significant trend	35817	5	2015 - 2019	24.03	0.07	23.26	0.05	0.6681

No detectable change in monthly average water temperature was observed at four locations. There was insufficient data to fit a model for five locations.

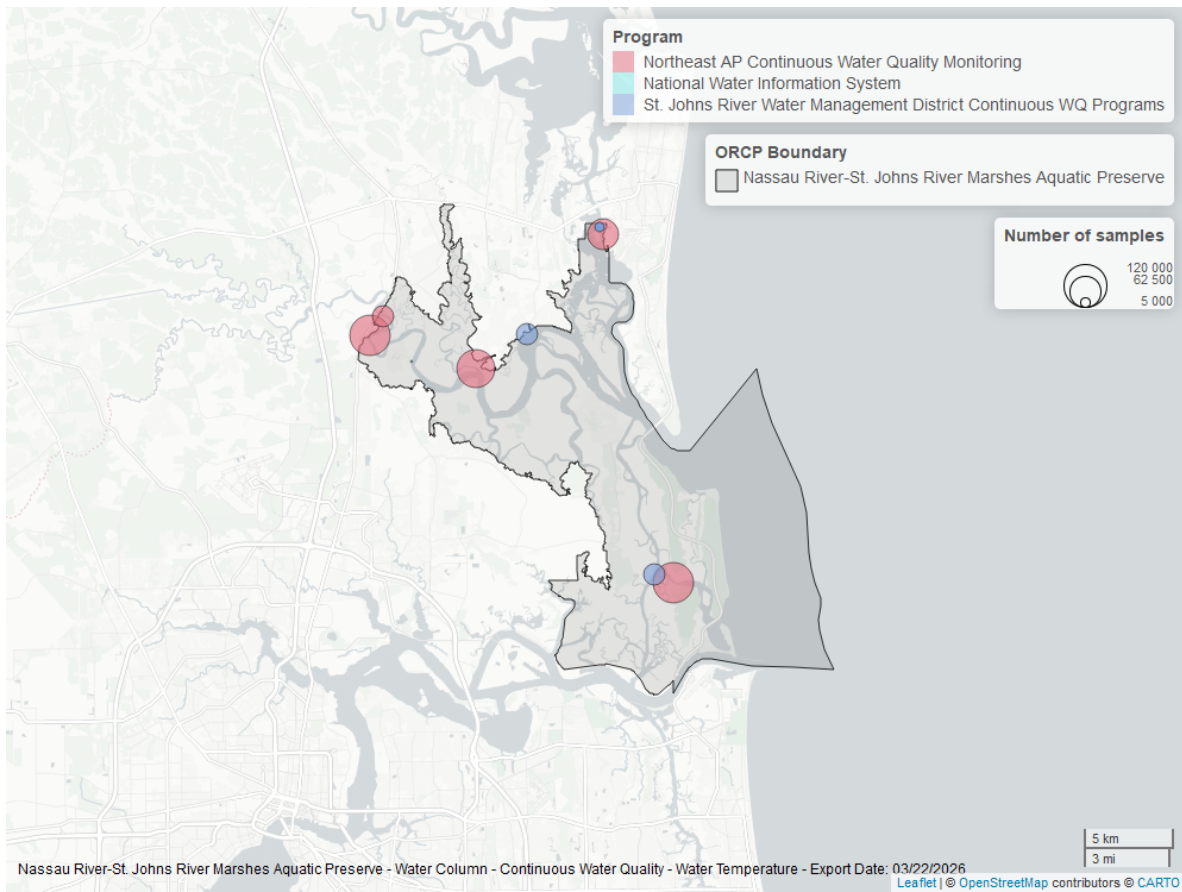


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

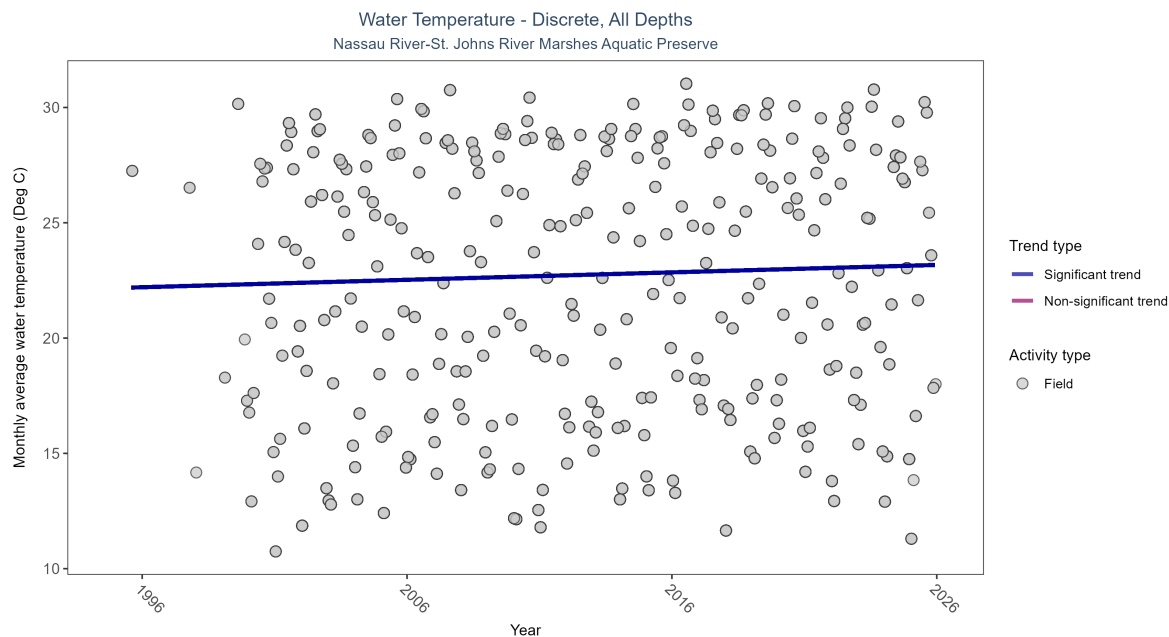


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	22846	30	1995 - 2025	23	0.11651	22.17189	0.03225	0.0041

Monthly average water temperature increased by 0.03°C per year.

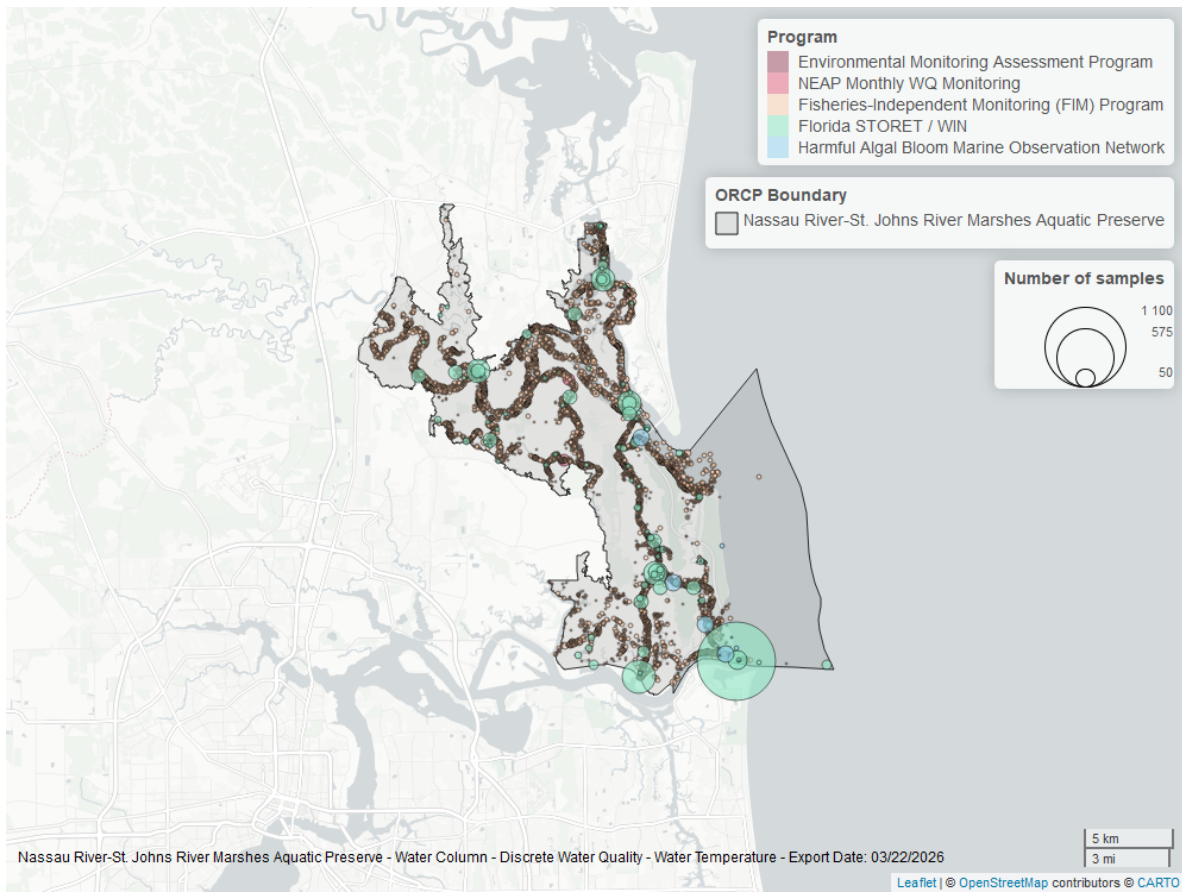


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

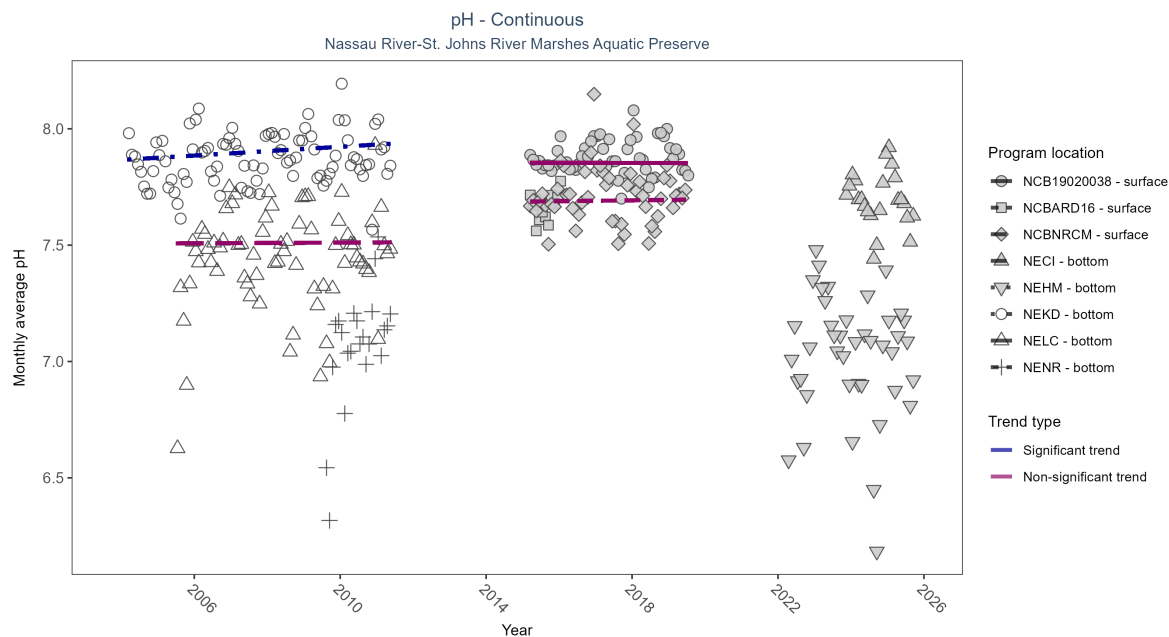


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data to calculate trend	60282	3	2023 - 2025	7.70	-	-	-	-
NEHM	Insufficient data to calculate trend	116425	4	2022 - 2025	7.10	-	-	-	-
NEKD	Significantly increasing trend	113471	8	2004 - 2011	7.90	0.19	7.87	0.01	0.0491
NELC	No significant trend	96488	7	2005 - 2011	7.50	0.02	7.51	0	0.9121
NENR	Insufficient data to calculate trend	31438	3	2009 - 2011	7.10	-	-	-	-
NCB19020038	No significant trend	34405	5	2015 - 2019	7.88	-0.01	7.85	0	1
NCBARD16	Insufficient data to calculate trend	6952	2	2015 - 2016	7.66	-	-	-	-
NCBNRCM	No significant trend	35819	5	2015 - 2019	7.72	-0.01	7.69	0	0.9317

At one program location, monthly average pH increased by 0.01 pH units per year. No detectable change in monthly average pH was observed at three locations. There was insufficient data to fit a model for four locations.

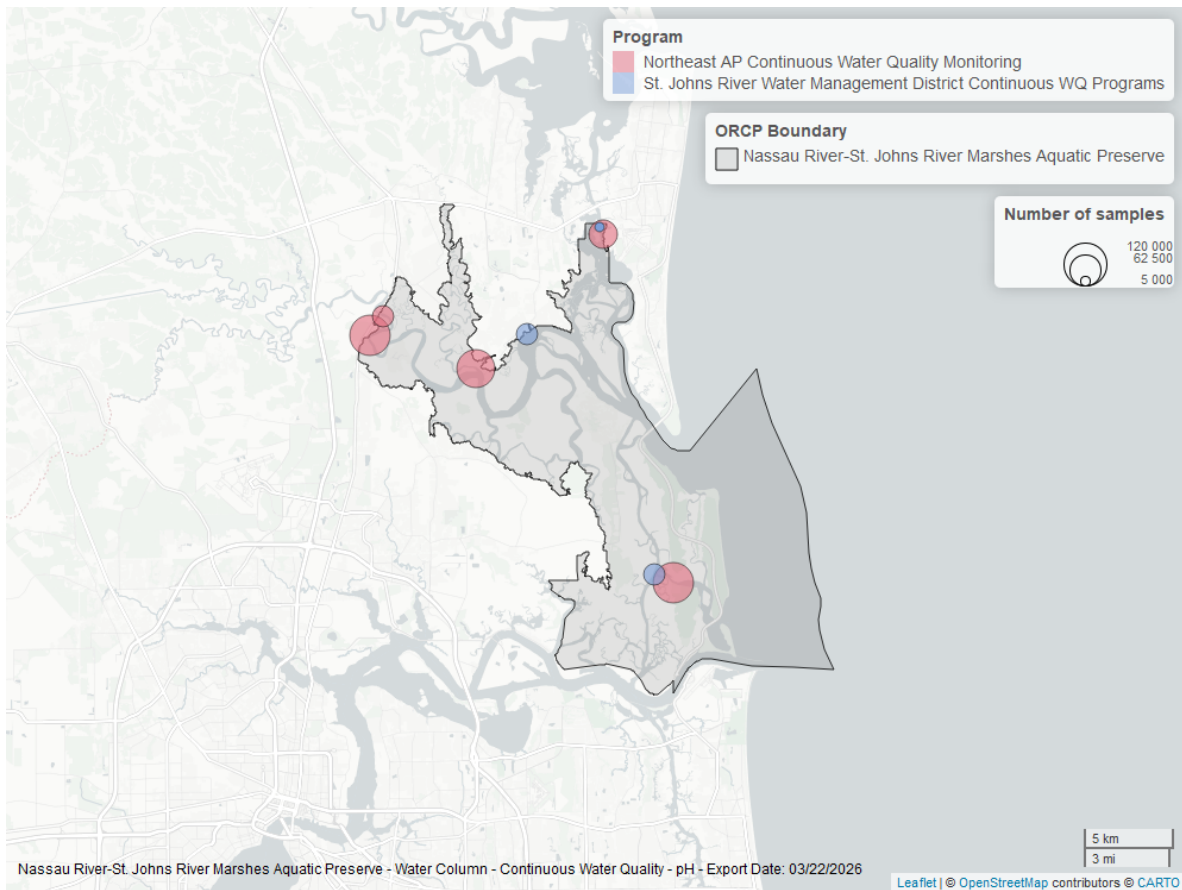


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

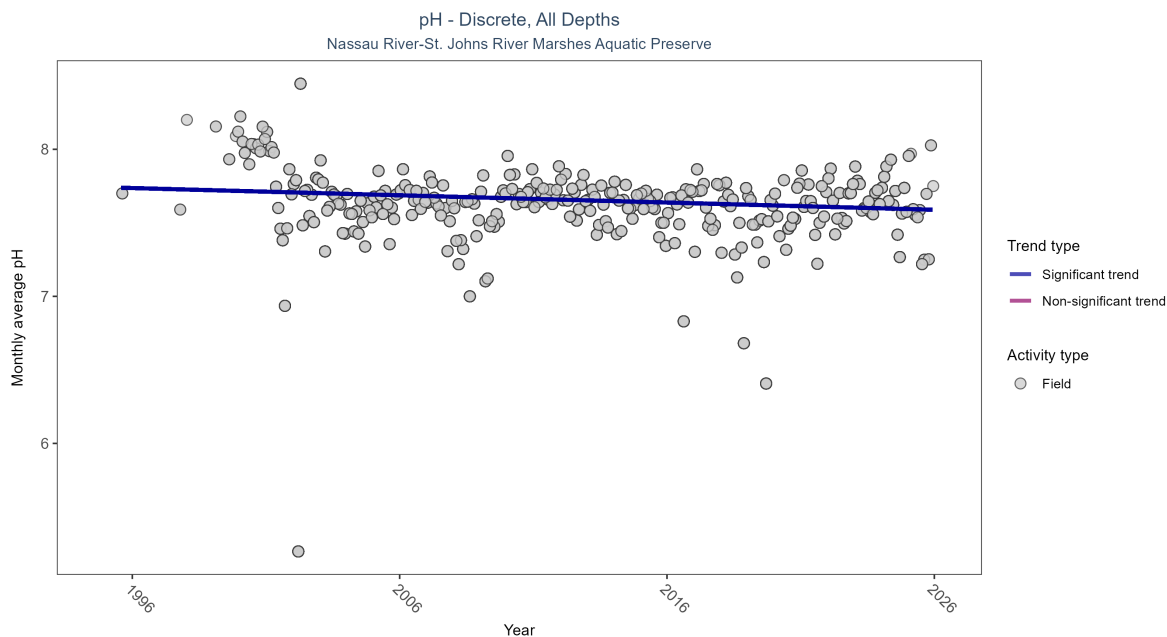


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	22611	30	1995 - 2025	7.6	-0.14003	7.74135	-0.00492	0.0005

Monthly average pH decreased by less than 0.01 pH units per year.

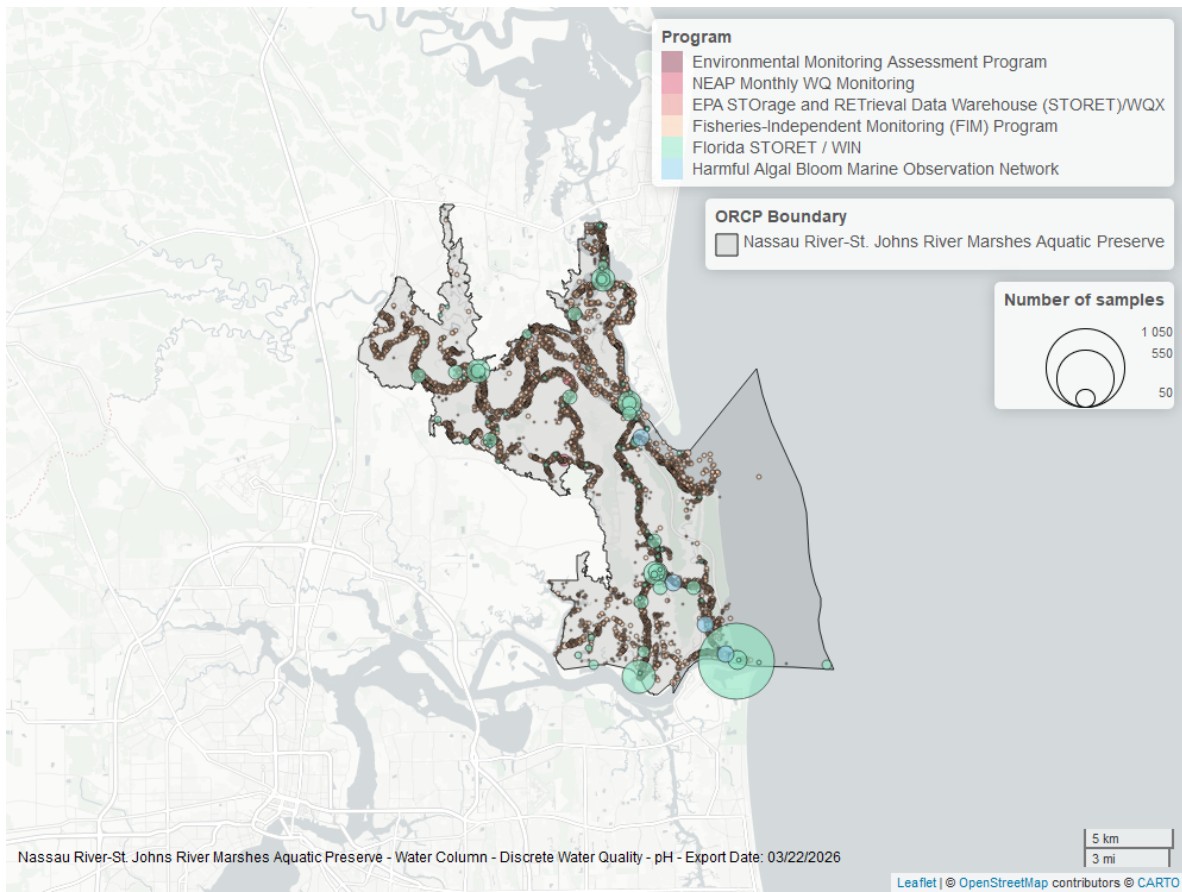


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

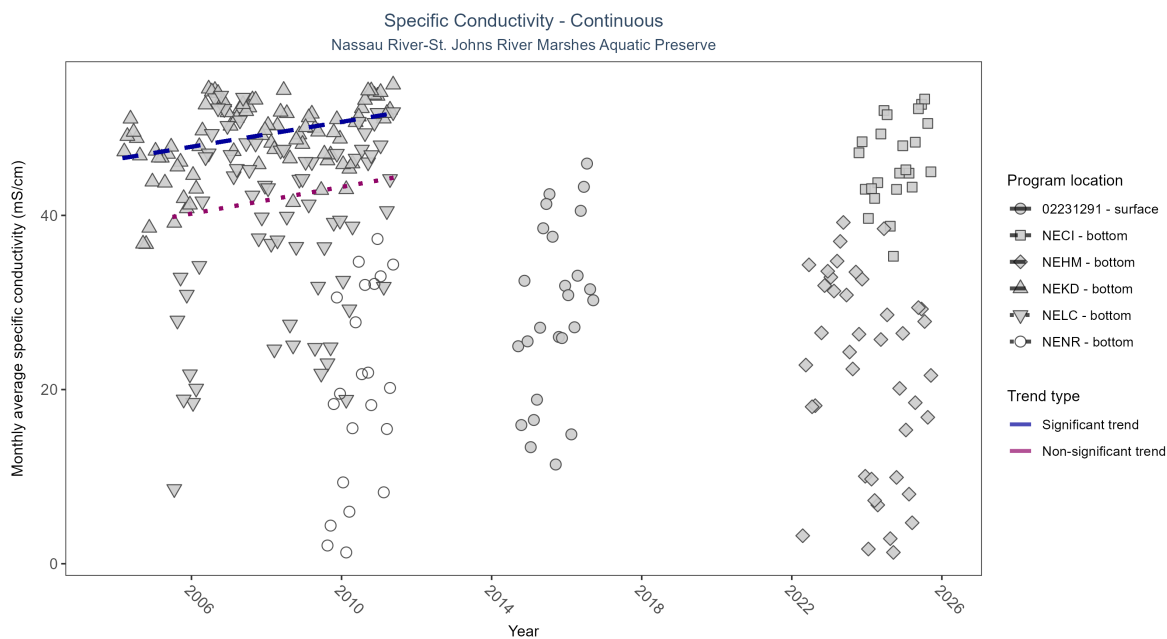


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
02231291	Insufficient data to calculate trend	682	3	2014 - 2016	30.40	-	-	-	-
NECI	Insufficient data to calculate trend	62231	3	2023 - 2025	45.94	-	-	-	-
NEHM	Insufficient data to calculate trend	111827	4	2022 - 2025	23.57	-	-	-	-
NEKD	Significantly increasing trend	118327	8	2004 - 2011	50.62	0.31	46.46	0.71	0.0006
NELC	No significant trend	100339	7	2005 - 2011	43.33	0.08	39.43	0.77	0.5078
NENR	Insufficient data to calculate trend	31438	3	2009 - 2011	20.04	-	-	-	-

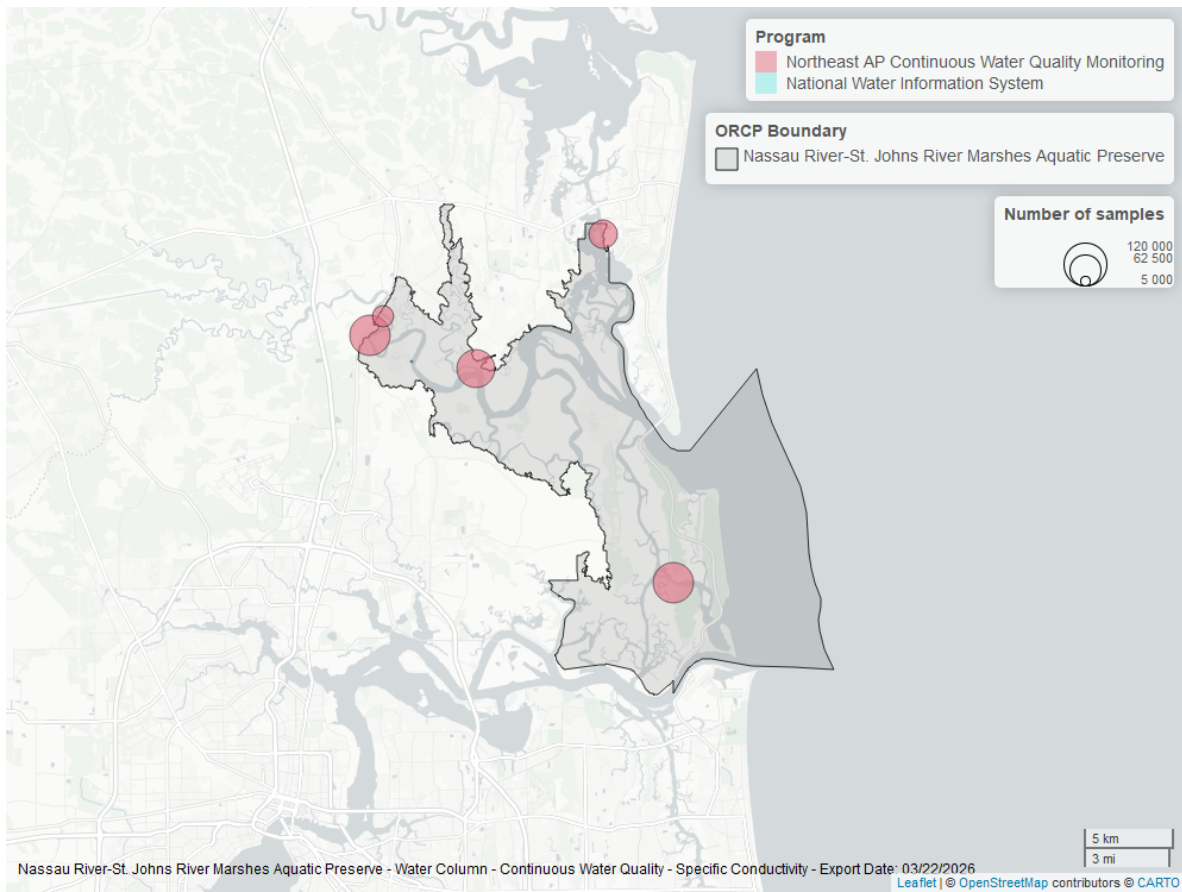


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

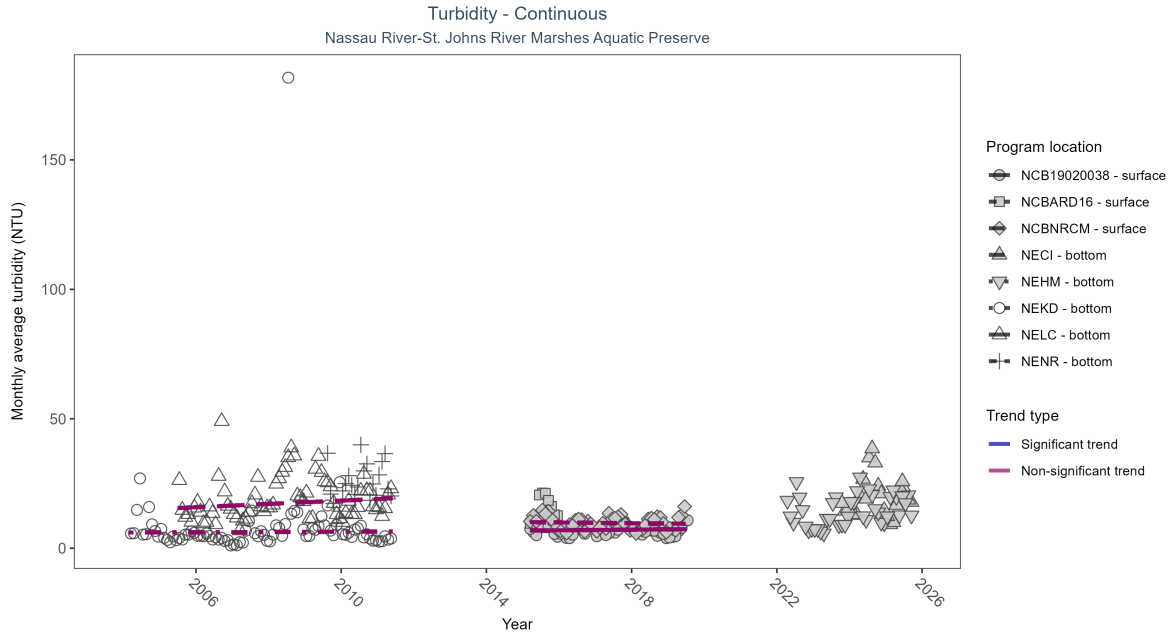


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
NECI	Insufficient data to calculate trend	60677	3	2023 - 2025	15.00	-	-	-	-
NEHM	Insufficient data to calculate trend	108736	4	2022 - 2025	12.00	-	-	-	-
NEKD	No significant trend	114181	8	2004 - 2011	4.00	0.02	6.06	0.05	0.8697
NELC	No significant trend	96153	7	2005 - 2011	15.00	0.16	15.18	0.64	0.1513
NENR	Insufficient data to calculate trend	31087	3	2009 - 2011	21.00	-	-	-	-
NCB19020038	No significant trend	31407	5	2015 - 2019	5.77	-0.01	6.79	0.12	0.7876
NCBARD16	Insufficient data to calculate trend	7385	2	2015 - 2016	12.89	-	-	-	-
NCBNRCM	No significant trend	34696	5	2015 - 2019	8.64	-0.1	10.09	-0.14	0.6001

No detectable change in monthly average turbidity was observed at four locations. There was insufficient data to fit a model for four locations.

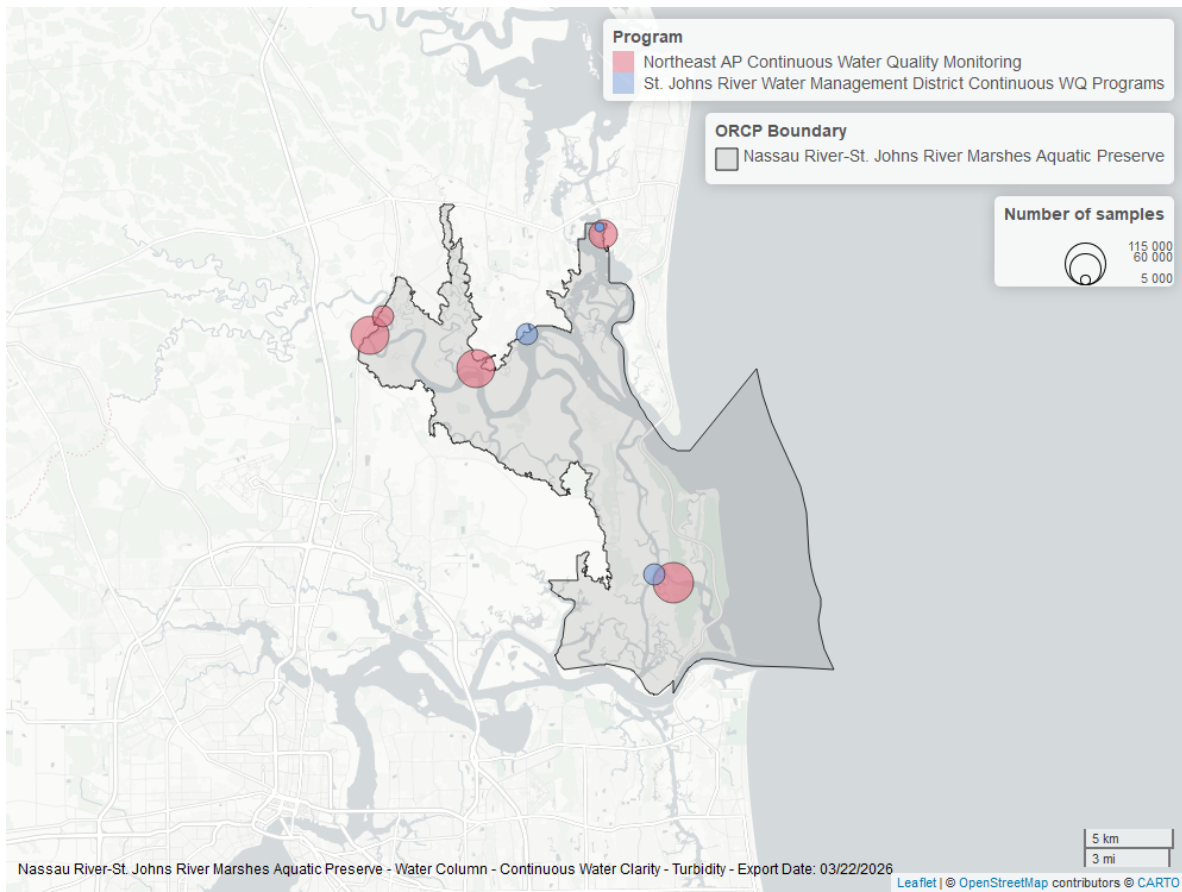


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

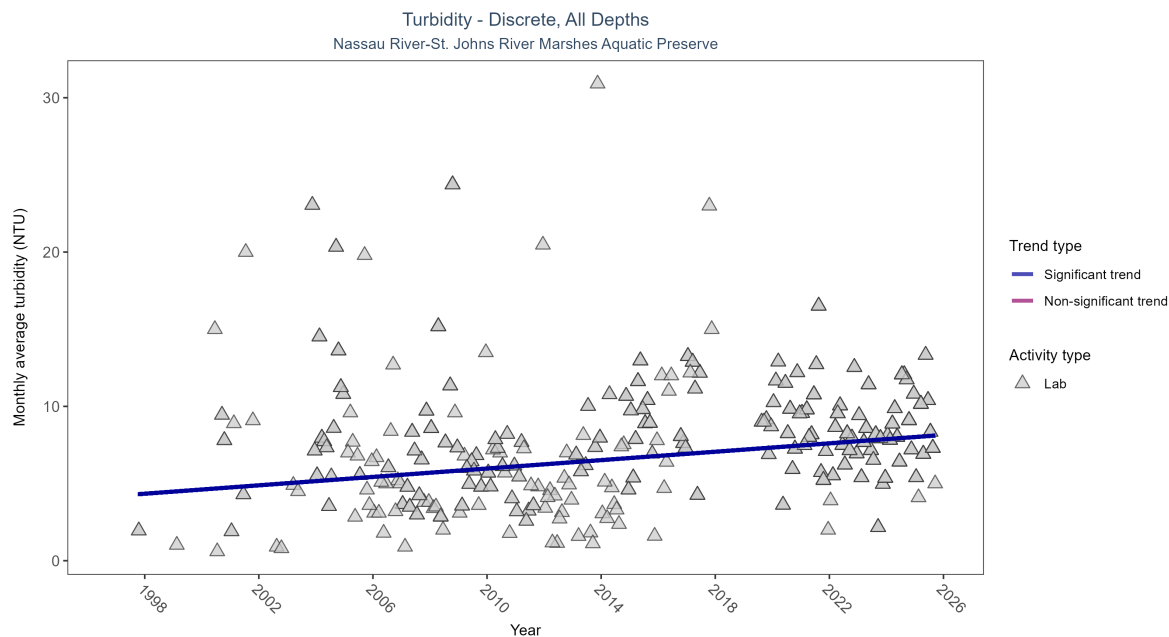


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	1098	27	1997 - 2025	7.10751	0.18191	4.20404	0.13605	0.0001

Monthly average turbidity increased by 0.14 NTU per year, indicating a decrease in water clarity.

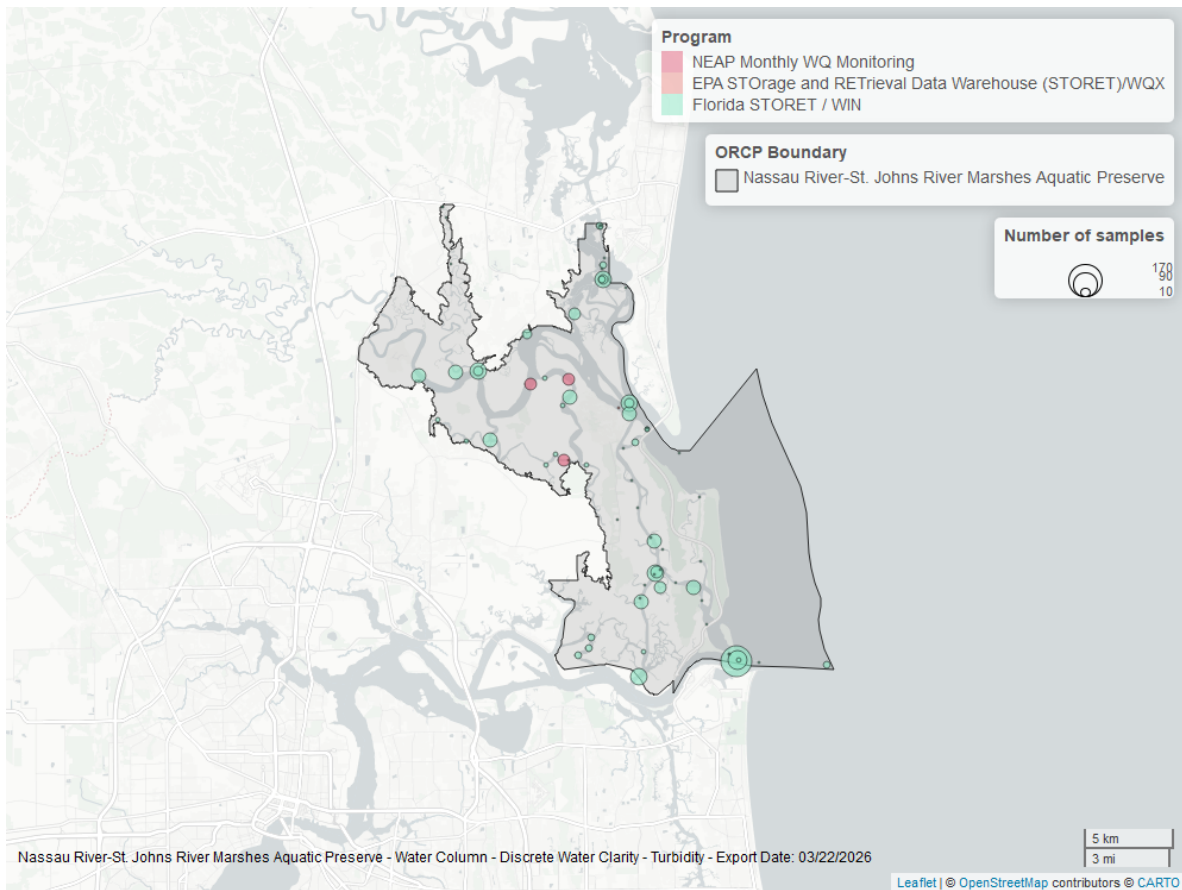


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

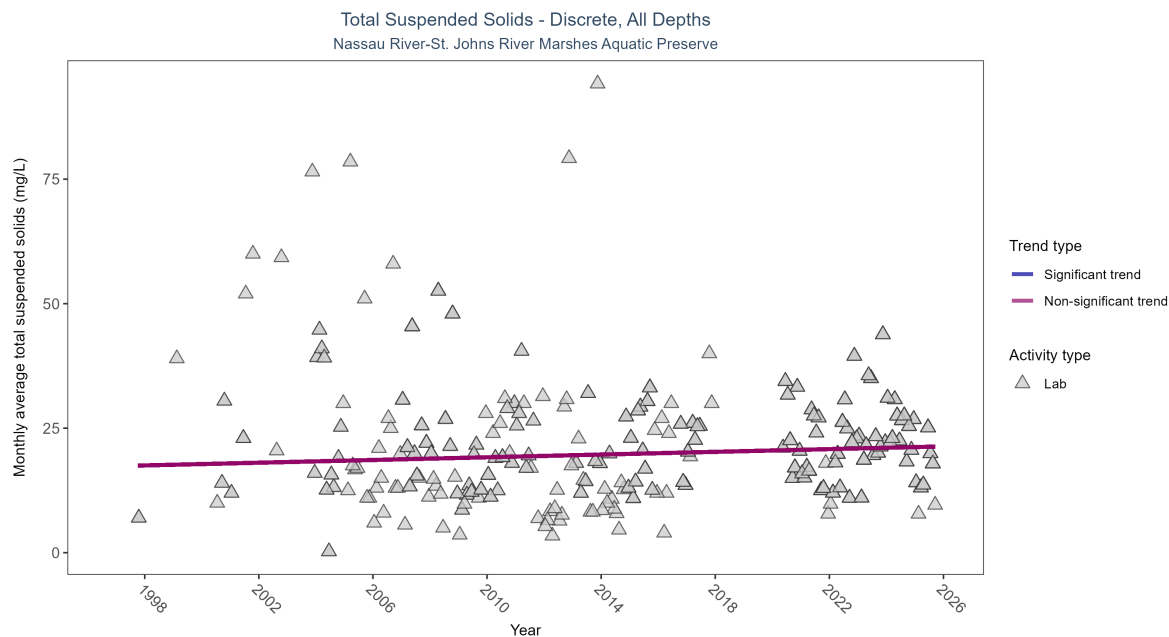


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	929	26	1997 - 2025	18	0.05882	17.38333	0.13636	0.2165

Total suspended solids showed no detectable trend between 1997 and 2025.

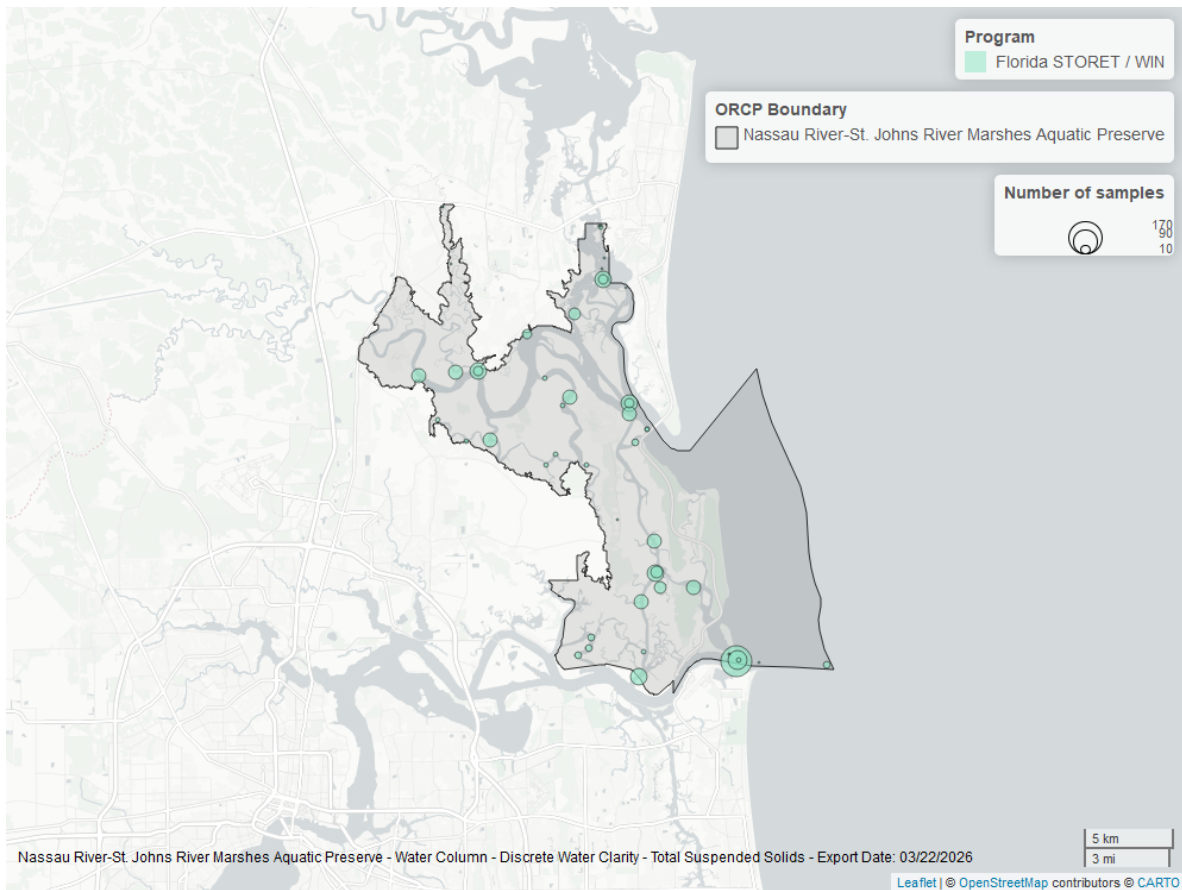


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

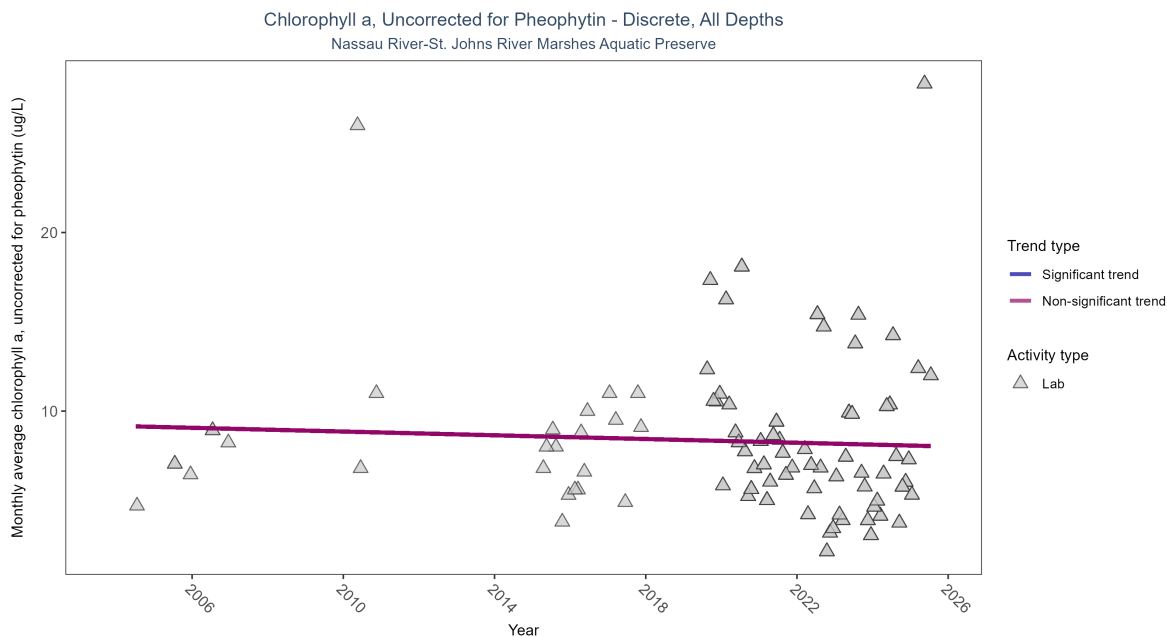


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	382	14	2004 - 2025	6.8	-0.10334	9.16876	-0.05231	0.6373

Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2004 and 2025.

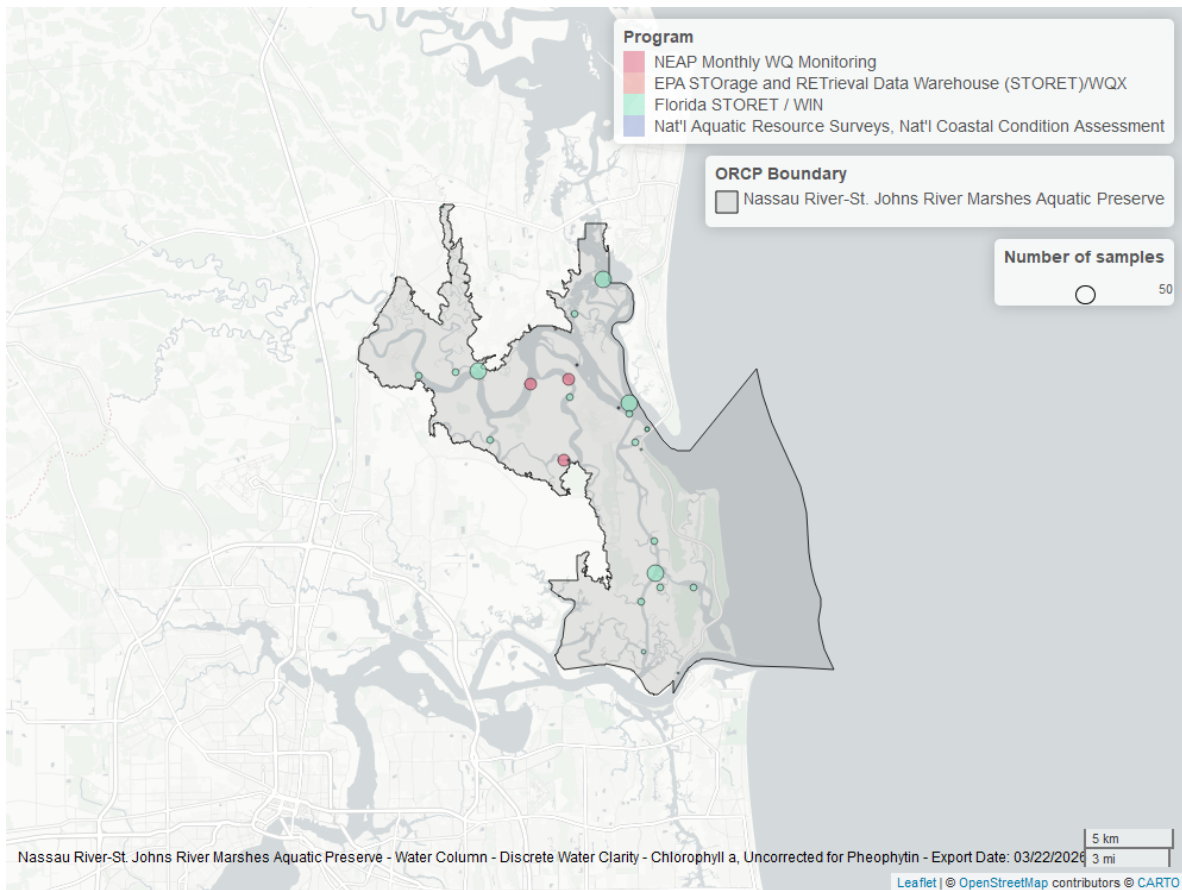


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

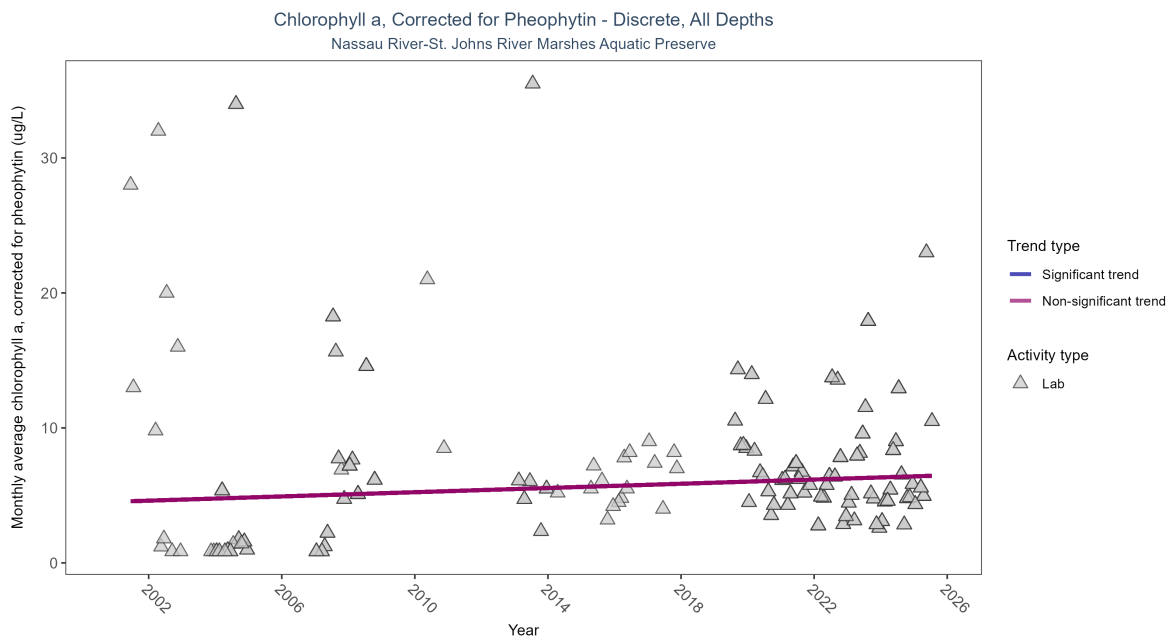


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	722	19	2001 - 2025	5.7	0.10558	4.53264	0.07828	0.1155

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2001 and 2025.

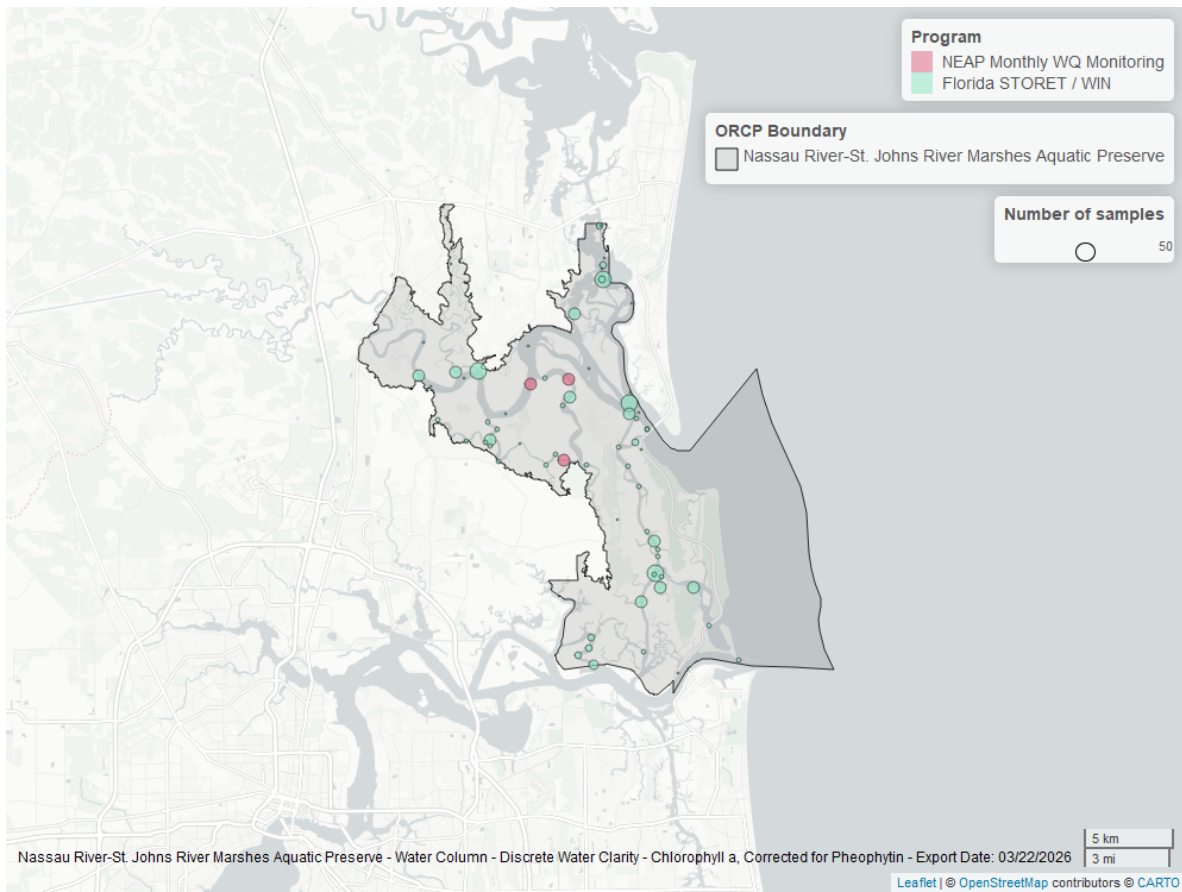


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

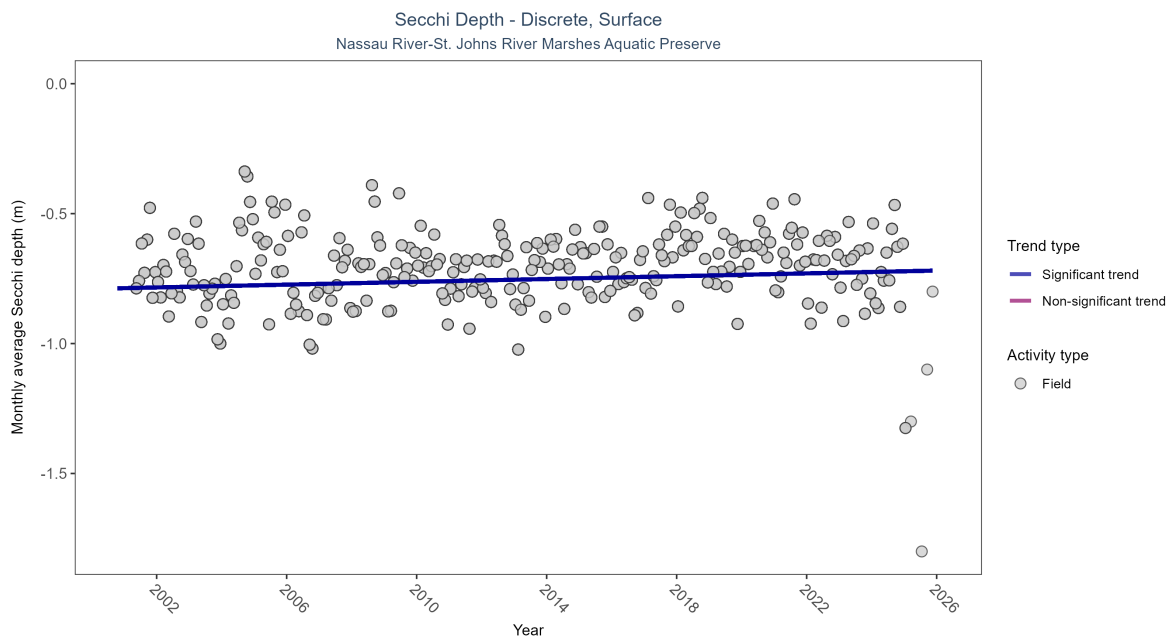


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	20403	26	2000 - 2025	-0.7	0.09425	-0.78956	0.00271	0.0281

Monthly average Secchi depth became shallower by less than 0.01 m per year, indicating a decrease in water clarity.

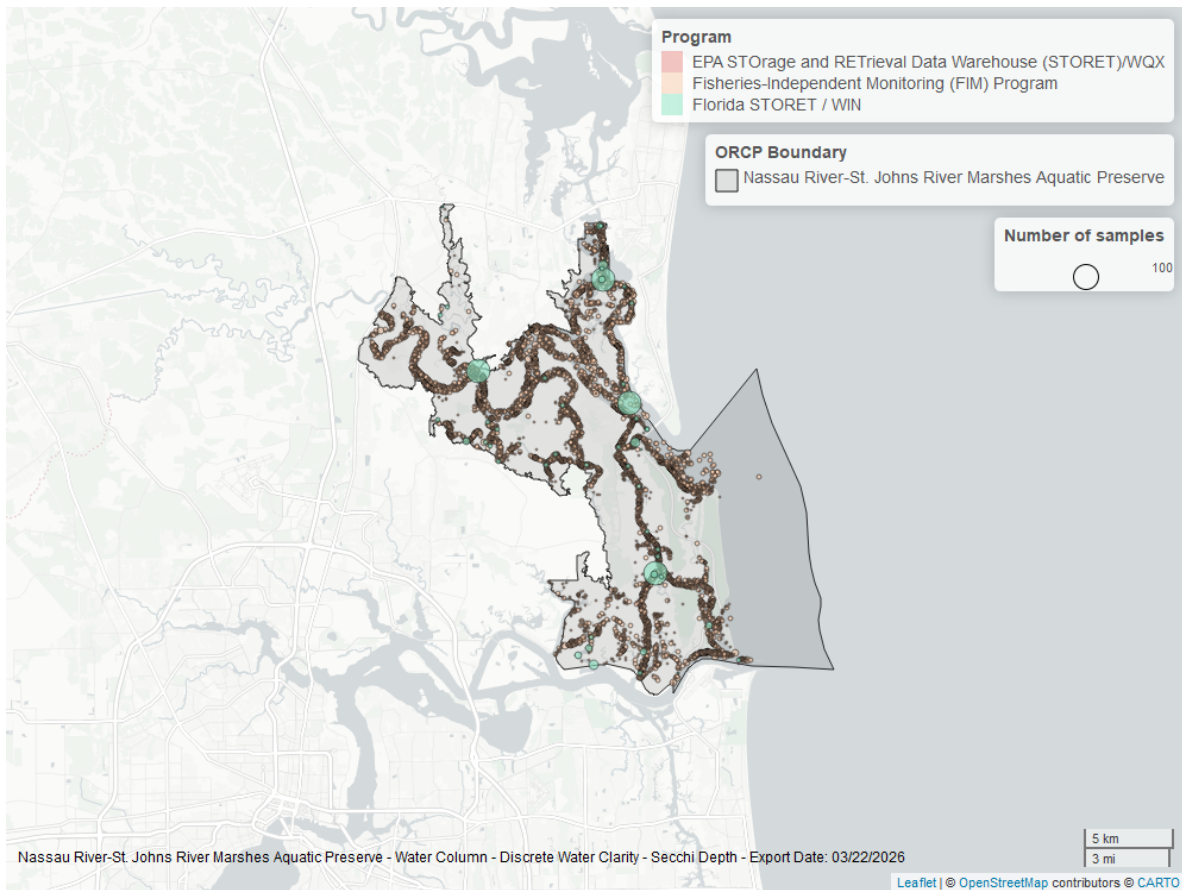


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

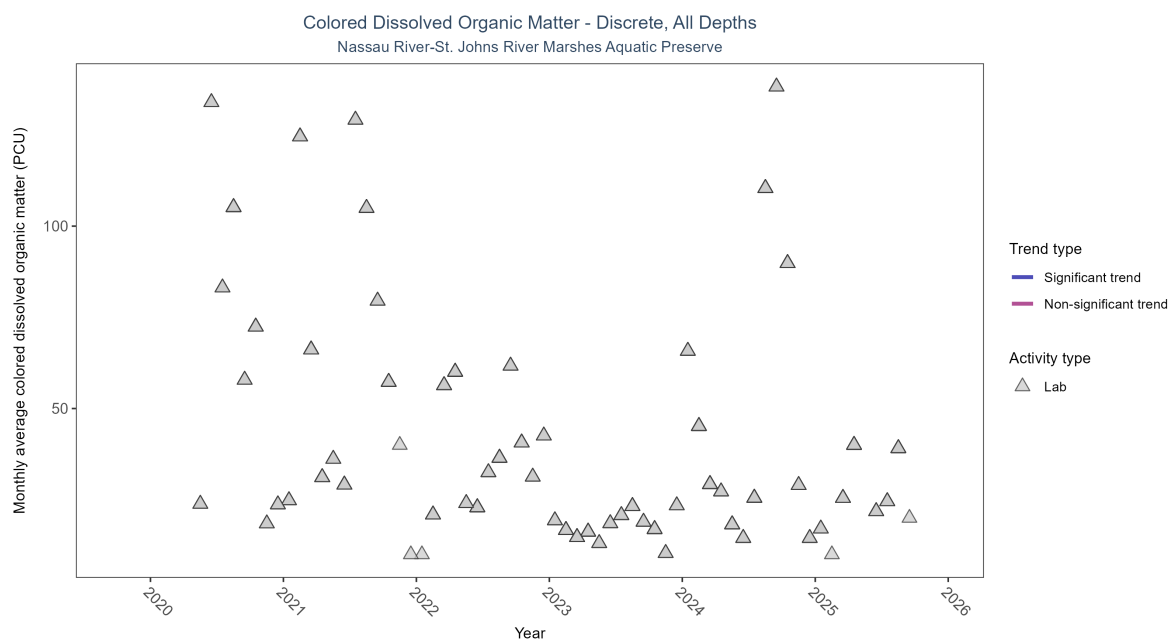


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data to calculate trend	551	6	2020 - 2025	26.27063	-	-	-	-

There was insufficient data to fit a model for colored dissolved organic matter.

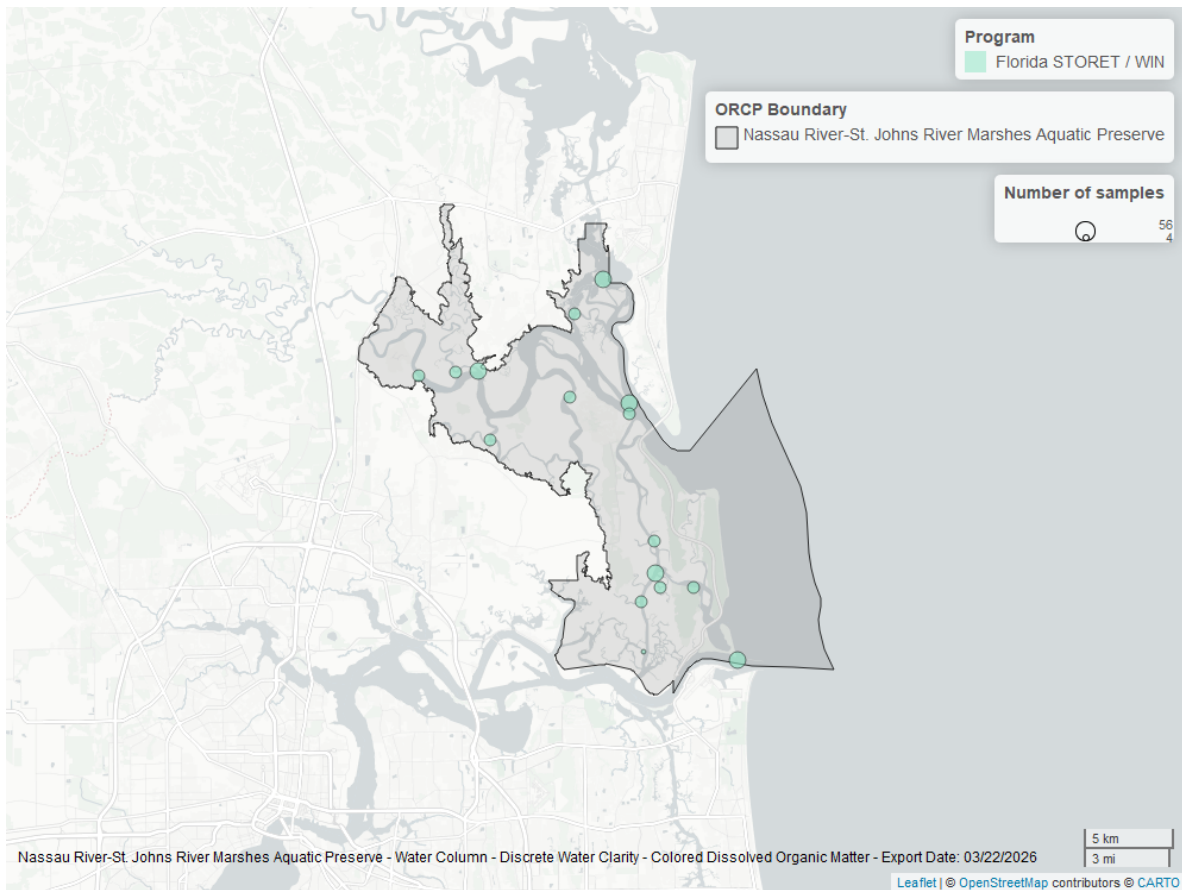


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Nassau River-St. Johns River Marshes Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.